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MILITARY SPECIFICATION

DRILLING MACHINE, WELL, ROTARY:

1,500-FOOT

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a well drilling machine, consisting of a power unit, mast, hoisting assembly, rotary table, mud pump, compressor and controls all mounted on a semitrailer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

L-P-378 - Plastic Sheet and Strip, Thin Gauge, Polyolefin.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3820

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O-C-114	- Calcium Hypochlorite, Technical.
T-R-605	- Rope, Manila and Sisal.
W-F-406	- Fittings for Cable, Power, Electrical and Conduit, Metal, Flexible.
W-F-408	- Fittings for Conduit, Metal, Rigid (Thick-Wall and Thin-Wall (EMT) Type).
W-F-1726/10	- Fuse, Cartridge, Class H, Nonrenewable, 600 volts, 101-200 Amperes.
W-S-610	- Splice Conductor.
HH-I-553	- Insulation Tape, Electrical (Rubber, Natural and Synthetic).
HH-I-595	- Insulation Tape, Electrical, Pressure Sensitive Adhesive, Plastic.
QQ-S-781	- Strapping, Steel, and Seals.
TT-C-494	- Coating Compound, Bituminous, Solvent Type, Acid Resistant.
TT-P-664	- Primer, Coating, Synthetic, Rust-Inhibiting, Lacquer-Resisting.
UU-S-48	- Sack, Shipping, Paper.
UU-T-81	- Tag, Shipping and Stock.
WW-C-581	- Conduit, Metal, Rigid, and Intermediate, and Coupling, Elbow, and Nipple, Electrical Conduits, Steel, Zinc Coated.
WW-P-521	- Pipe Fittings, Flange Fittings, and Flange, Steel and Malleable Iron (Threaded and Butt Welded) Class 150.
WW-U-531	- Unions, Pipe, Steel or Malleable Iron, Threaded Connection, 150 Lb and 250 Lb.
ZZ-H-561	- Hose, Rubber, and Hose Assemblies, Rubber, Smooth Bore, Water Suction and Discharge.
GGG-T-581	- Taps, Dies, Diestocks, and Threading Set---Pipe Threading, Hand.
GGG-V-415	- Vise, Pipe.
PPP-B-601	- Boxes, Wood, Cleated Plywood.
PPP-B-636	- Boxes, Shipping Fiberboard.
PPP-P-40	- Packaging and Packing of Hand Tools.

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MIL-V-3	- Valve, Fittings, and Flanges (Except for Systems Indicated Herein), Packaging of.
MIL-C-104	- Crates, Wood: Lumber and Plywood Sheathed, Nailed and Bolted.

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MIL-P-116	- Preservation, Methods of.
MIL-B-121	- Barrier Material, Greaseproofed, Waterproofed, Flexible.
MIL-S-416	- Strainer, Well, Water, Wire Wound.
MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-C-3131	- Cordage, Packaging of.
MIL-F-3541	- Fitting, Lubrication.
MIL-C-3600	- Compressor, Rotary, Power-Driven; and Compressors, Air and Gas (Except Oxygen), Packaging of.
MIL-G-3859	- Grease Gun, Hand Operated, Lever, Push and Screw Type.
MIL-A-8421	- Air Transportability Requirements, General Specification for.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-G-10924	- Grease, Automotive and Artillery.
MIL-C-12000	- Cable, Cord, and Wire, Electric Packaging of.
MIL-E-17555	- Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-L-21260	- Lubricating Oil, Internal Combustion Engine, Preservative and Break-In.
MIL-T-22085	- Tape, Adhesive, Preservation and Sealing.
MIL-D-23003	- Deck Covering Compound, Nonslip Rollable.
MIL-G-23827	- Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-G-28554	- Generator Set, Mobile Electric Power and Supplemental Equipment, Packaging of.
MIL-P-46093	- Primer Coating, Synthetic (For Brake Drums).
MIL-A-46153	- Antifreeze, Ethylene Glycol, Inhibited, Heavy Duty, Single Package.
MIL-F-52723	- Filter Element, Hydraulic, Disposable.
MIL-C-62122	- Cable Assembly, Slave, Electrical Connectors, Plug and Receptacle.
MIL-S-81733	- Sealing and Coating Compound, Corrosion Inhibitive.

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STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	- Identification Marking of US Military Property.
MIL-STD-163	- Steel Mill Products, Preparation for Shipment and Storage.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281	- Automobiles, Trucks, Truck-Tractors, Trailers and Trailer Dollies; Preservation and Packaging of.
MIL-STD-461	- Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1186	- Cushioning, Anchoring, Bracing, Blocking and Waterproofing, With Appropriate Test Methods.
MIL-STD-1188	- Commercial Packaging of Supplies and Equipment.
MIL-STD-1410	- Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472	- Human Engineering Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.
MS27183	- Washer, Flat-Round, Steel, Cadmium Plated, General Purpose.
MS35338	- Washer, Lock-spring, Helical, Regular (Medium) Series.
MS51336	- Lunette-Coupler, Drawbar, Ring.
MS51953	- Nipple, Pipe, Standard Steel.
MS51967	- Nut, Plain, Hexagon-carbon Steel, Cadmium Plated, UNC-2B.
MS90725	- Screw, Cap, Hexagon Head (Finished Hexagon Bolt), Steel, Grade 5, Cadmium Plated, UNC-2A.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DEPARTMENT OF TRANSPORTATION (DOT)

Federal Motor Vehicle Safety Standards (FMVSS)
Federal Motor Carrier Safety Regulations (FMCSR).

(Applications for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

DRAWINGS

ME

13218E0479	- Adapter, Straight.
13226E1147	- Frame Assembly, Reduced Voltage Starter.
13226E1148	- Frame, Skid.
13226E1149	- Cap, Well Head.
13226E1150	- Clamp, Pipe.
13226E1230	- Frame, Mounting.
13226E1231	- Rod.

(Copies of specifications, standards, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI/ASME - Boiler and Pressure Vessel Code, Section VIII, Unfired Pressure Vessels.

ANSI/ASME - Boiler and Pressure Vessel Code, Section IX, Welding Qualifications.

ANSI/AWS D1.1 - Structural Welding Code, Section 5, Qualification.

B16.5 - Steel Pipe Flanges and Flanged Fittings, Including Ratings for Class 150, 300, 400, 600, 900, 1500 and 2500.

B16.21 - Nonmetallic Flat Gaskets for Pipe Flanges.

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(Application for copies should be addressed to the American National Standards Institute, Inc., 1430 Broadway, New York, NY 10018.)

AMERICAN PETROLEUM INSTITUTE (API)

Specification 5L - Specification for Line Pipe.

(Application for copies should be addressed to the American Petroleum Institute, 1801 K Street, NW, Washington, DC 20037.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A120 - Pipe, Steel, Black and Hot-Dipped Zinc-Coated (Galvanized) Welded and Seamless, for Ordinary Uses.
C150 - Portland Cement.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN TRUCKING ASSOCIATIONS (ATA)

E-3 - Electrical Lighting and Wiring for Commercial Vehicle Bodies.

(Application for copies should be addressed to the American Trucking Associations, Inc., 1616 F Street NW, Washington, DC 20036.)

NATIONAL FLUID POWER ASSOCIATION (NFLDP)

Method T3.10.8.8 - Multi-Pass Method for Evaluating the Filtration Performance of a Fine Hydraulic Fluid Power Filter Element.

(Application for copies should be addressed to the National Fluid Power Association, 3333 North Mayfield Road, Milwaukee, WI 53222.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15086.)

MANUFACTURERS STANDARDIZATION SOCIETY OF THE VALVE AND FITTINGS INDUSTRY, INC (MSS)

MSS-SP80 Bronze Gate, Globe, Angle and Check Valves

(Application for copies should be addressed to the Manufacturers Standardization Society of the Valve and Fittings Industry, Inc., 5203 Leesburg Pike, Suite 502, Falls Church, VA 22041.)

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TIRE AND RIM ASSOCIATION (TRA)

Yearbook.
Supplementary Service Data Book.

(Application for copies should be addressed to the Tire and Rim Association, Inc., 3200 West Market Street, Akron, OH 44313.)

TRUCK-TRAILER MANUFACTURERS ASSOCIATION (TTMA)

TTMA Recommended Practices.

(Application for copies should be addressed to the Truck-Trailer Manufacturers Association, Inc., 2430 Pennsylvania Avenue, NW, Washington, DC 20037.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take preference.

3. REQUIREMENTS

3.1 Description. The drilling machine shall consist of a mast, drawworks, drilling machinery, rotary table, mud pump, and air compressor, driven by a diesel engine and power transmission components, all permanently installed on a tandem axle semitrailer.

3.1.1 Weights and dimensions. The gross vehicle weight (gvw) shall include the operating machinery equipment and accessories installed on the semitrailer and prepared for over-the-highway transport. The physical characteristics of the drilling machine shall be as specified in table I, with the drilling machine on level ground and with the inflation pressure of the tires as recommended in the TRA Yearbook for gvw operation at highway speed. Published component and vehicle ratings shall not be increased to meet the requirements of this specification.

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TABLE I. Weights and dimensions.

(a) Gvw, max., lb.	46,000
(b) Fifth wheel weight, max., lb.	15,000
(c) Axle load, max., lb (each)	17,000
(d) Overall length, max., ft. (less dolly)	45
(e) Overall height, max., ft.	12.5
(f) Overall width, max., ft. (exclusive of clearance lights)	8
(g) Ground clearance, min., in. (SAE J894)	9.5
(h) Ground clearance, min., in. (exclusive of rear suspension)	14
(i) Landing wheel clearance, radius, min., in. ("LWC" of SAE J701)	86
(j) Swing radius, max., in. ("SR" of SAE J701)	58
(k) Kingpin setting, range, in. ("KP" of SAE J701)	18 through 36
(l) Upper fifth wheel height, in.	55 $\pm$1
(m) Angle of departure, min., degrees (SAE J894)	28

3.1.2 Compatibility. The drilling machine shall be compatible for towing behind a 10-ton, truck tractor.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3).

3.3 Material. Material shall be as specified herein and on the drawings. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification (see 6.9).

3.3.1 Material deterioration and control. The drilling machine shall be fabricated from compatible materials, inherently corrosion and deterioration resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage environment to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals, as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by an insulating barrier such as a corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

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3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request, to the contracting officer or the designated representative.

3.3.2 Rubber products. Rubber products furnished shall not exceed 8 quarters in age from the date of cure of the rubber to the date of acceptance of the drilling machine by the Government.

3.4 Threads. All screw threads shall conform to FED-STD-H28, and SAE standards.

3.5 Safety. The safety characteristics of the drilling machine shall meet the following minimum requirements:

a. Provide safety or protective devices, which do not impair the operating functional requirements, or which furnish prominent warnings.

b. Provide guarding, enclosing, and insulating features for exposed components and systems which are subject to high temperatures, high pressure, electric actuation, or which are inherently hazardous.

c. Provide nonskid footing on all walk and step surfaces.

d. The engine exhaust and other discharges from the equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator.

e. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine.

3.6 Human factors engineering. The drilling machine shall conform to MIL-STD-1472 (see 4.5.2.4).

3.6.1 Noise limits. The noise of the drilling machine (excluding horns and warning alarms) shall not exceed the limits specified in MIL-STD-1474, category D for steady-state noise when tested in accordance with 4.5.2.5.1. The sound level shall comply with the exterior noise requirements of MIL-STD-1474 when tested in accordance with 4.5.2.5.2 (aural nondetectability and impulse noise do not apply). The precautions of MIL-STD-1474 (see 6.8) shall be provided if the sound level exceeds 85 dB(A) when tested in accordance with 4.5.2.5.3. Hearing protectors shall not be supplied with the drilling machine. Hazard signs, when required, shall state "CAUTION, HEARING PROTECTION REQUIRED". Plates shall be in accordance with MIL-P-514, type III, composition C of type I, grade A, class 1 material.

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3.7 Reliability. The specified mean-time-between-failure (MTBF) shall be 100 hours when the drilling machine is tested as specified in 4.5.2.6.

3.8 Maintainability.

3.8.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.6 and 6.7), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.5.2.7.

3.8.2 Maintenance ratio. All maintenance performed during testing specified in 4.5.2.7 shall be accomplished by not more than two persons using the tools and equipment available to the using organization. The ratio of man-hours of maintenance to total operating hours shall not exceed 0.15. A maintenance schedule shall be furnished prior to the start of any testing.

3.9 Ease of maintenance. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement with minimum removal of other major assemblies and installed attachments. Covers, plates, or panels which must be removed for component adjustment, repair, replacement, or maintenance shall be equipped with clinch- or weld-nuts, threaded inserts, quick-disconnect fasteners, or threaded base members. Removal of all fasteners shall be accomplished by one mechanic without requiring the assistance of a helper. The configuration of the drilling machine shall provide maximum accessibility for removal of engine and power transmission components. Where incorrect installation of an item could cause malfunction of that item or of the system in which it is installed, an unsymmetrical mounting means shall be provided or the mating parts shall be doweled or matchmarked. Ease of maintenance provisions shall incorporate features insuring operating tool and lubricant component clearances in accordance with SAE J925 and shall facilitate maintenance and servicing operations, including checking and adding of engine oil and coolant.

3.9.1 Enclosures with integral reservoir of lubricant. Enclosures such as gearcases and transmission housings that contain a reservoir of lubricants for the lubrication of the parts enclosed shall be equipped with dipsticks, check plugs not less than 1/2-inch pipe size, or sight gages to determine the level of the lubricant. Each enclosure shall be equipped with a means for filling the enclosure with a lubricant and for draining. The drain shall be fitted with a drain plug. The drain shall be located so that removal of the plug will result in complete drainage of the lubricant from the enclosure. Drainage shall be clear of other equipment or framing when the drilling machine is in its normal operating position. Integral tubes or troughs may be used to convey the lubricant from the drain to a container. The drain plugs, the filling means, and the lubricant-level dipsticks or holes shall be accessible without disturbance of accessories or parts.

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3.10 Performance. The drilling machine shall drill through soil and rock formations and shall set 20-foot lengths of casing conforming to API Spec. 5L (line pipe), in accordance with the minimum performance requirements specified in table II.

TABLE II. Performance requirements (minimum).

Casing (line pipe) water well reamed and drifted threaded and coupled				
Bit size (inches)	Outside diameter (inches)		Wall thickness (inches)	Size Nominal (inches)
	Pipe	Coupling		
13-3/4	10.75	11.75	.365	10
9-7/8	8.625	9.625	.322	8
7-7/8	6.625	7.390	.280	6

Table II specifies minimum performance requirements for the drilling machine when using either mud rotary or air rotary techniques. In addition, the drilling machine shall be capable of drilling a 6-inch hole through rock to a depth of not less than 1,500 feet using the down-the-hole rotary percussion technique. The drilling machine shall also have an air/form-circulation drilling capability.

3.10.1 Mobility. The entire drilling machine shall be capable of being towed at speeds as great as 10 miles per hour (mph) over unpaved roads, and over reasonably hard, uneven terrain; and at speeds as great as 55 mph over improved roads.

3.11 Diesel engine. For the purpose of selecting a diesel engine, the drilling machine shall be classified as a class II end item in accordance with MIL-STD-1410. The engine, including all systems, components, and accessories, shall conform to MIL-STD-1410, class II, except as otherwise specified herein. The engine shall be furnished complete with the following:

a. A fuel tank of sufficient capacity for 12 hours rated load operation.

b. A 24-volt (V) electrical cranking system. Two batteries shall be furnished. Each battery shall be 12 V potential, lead acid, negative ground, and have a capacity of not less than 200 ampere-hours at the 20-hour rate. Each battery shall be shipped, charged and dry. Electrolyte shall not be furnished.

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c. A battery charging alternator with a rating of not less than 50 amperes.

d. A two-stage, dry type, intake, air cleaner with service indicator.

e. A waterproof slave connector receptacle, conforming to MIL-C-62122.

f. Safety control devices of the visual or audible type for high coolant temperature and low lubricating oil pressure.

g. Hood and side doors.

h. A mechanical, liquid quantity, fuel indicator located at the tank.

i. Hour meter.

j. A limiting speed governor.

k. Mufflers.

3.11.1 Engine operation requirements. The engine operation requirements and maximum engine operating limits shall be in accordance with MIL-STD-1410 under any of the requirements and conditions specified herein.

3.11.2 Power and speed rating. The power and speed rating of the engine shall be such that operation of the drilling machine at the conditions specified herein will not require horsepower in excess of the continuous brake horsepower rating of the engine for the applicable governed speed.

3.11.3 Engine control systems. The drilling station shall be equipped with, but not necessarily limited to, the following engine instruments and controls:

a. Lubricating oil pressure indicator.

b. Cooling liquid temperature indicator.

c. Tachometer.

d. Battery-alternator indicator.

e. Manual speed control.

f. Emergency shutdown device.

g. Normal shutdown device.

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3.12 Transfer case. A fully enclosed, oil bath lubricated, chain and sprocket transfer case shall be furnished to distribute engine power. Chains shall be multiple strand, precision roller type. Sprockets and shafts shall be heat-treated alloy steel, mounted on heavy duty anti-friction type bearings. A method of adjusting chains to compensate for wear shall be provided and be accessible. Clutches shall be furnished for, but not necessarily limited to, the rotary assembly, fluid circulation system and hoisting equipment.

3.13 Drilling machinery. The drilling machinery shall be installed on the semitrailer frame to withstand all operational and transport forces specified herein without misalinement, malfunction, damage, or permanent deformation. A raised pattern deck plate not less than 0.125 inch thick shall cover the entire open area of the top surface on the framework. The deck plate shall be secured to the frame or subframe. Solid metal deck plates shall be painted with a non-skid, abrasive type deck paint, conforming to MIL-D-23003, type II. The decking shall support not less than 400 pounds per square foot, imposed on any 1-square foot at a time. The drill frame shall be equipped with a hinged or detachable driller's and helper's platform. A 4-1/2-inch swivel base vise shall be mounted on the deck area of the semitrailer. A deadline anchor and weight indicator (40,000 pounds) shall be furnished.

3.13.1 Mast. The drilling machine shall be equipped with a mast constructed of all electrically welded structural steel or steel tubing. The mast shall sustain a gross capacity of 40,000 pounds and shall be sufficient height to accommodate range 1, nominal 20-foot sections of drill pipe and 21-foot sections of casing for all operations of drilling when using any combination of accessories furnished. A structural steel ladder shall be permanently affixed to the mast to permit accessibility to any part of the mast. The mast shall be raised and lowered by one or two hydraulic cylinders. The cylinders shall have chrome plated cylinder rods and shall have safety devices to prevent an uncontrolled lowering of the mast. The mast shall be equipped with kelly bar holders to attach kelly bar and kelly swivel for transit. The mast shall be completely supported by the main frame and a locking device to lock the mast in its drilling position shall be furnished.

3.13.1.1 Mast supports and steady jacks. When the mast is not completely supported by the main frame, front mast supports shall be provided. A locking mechanism shall be provided to rigidly fasten and to mechanically lock the mast in an upright position. When the mast is lowered and in a stowed position it shall not interfere with the towing vehicle.

3.13.2 Crown block. The mast shall be fitted with a crown block assembly. The crown block shall have a sufficient number of sheaves to permit either 2- or 3-part reeving of the main hoist and 2-part reeving on the sand reel hoist. Hoisting and sand line sheaves with detachable guards shall be furnished. An eye type anchor shall be attached to the underside of the crown block. A rope sheave to accommodate 1-inch manila rope shall be located in the mast for use with the cathead.

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3.13.3 Drawworks. Drawworks shall consist of a main hoist of the double drum type, sand-reel hoist, cathead, and accessories. Component parts shall be interchangeable wherever possible. The drawworks shall be driven by a spiral bevel ring gear and pinion or by chain and sprockets through the power transmission. When chain and sprockets are indirectly driven, power shall pass through a bevel gearcase. The bevel gearcase output shaft shall be equipped with a drive sprocket. Shafts, sprockets, and chains shall be heat-treated, alloy steel. Chains shall be multiple strand, precision roller type. The bevel ring gear and pinion of the two drives shall be hardened alloy steel with ground, spiral cut teeth. Both cases shall be fitted with oil drain and filler plugs located to assure correct servicing. All shafts of the hoisting equipment shall be mounted on heavy duty ball or roller bearings. Drawworks controls shall be located at the drilling station.

3.13.3.1 Main hoist. The main hoist shall be double drum type. Each drum shall be equipped with a friction clutch and brake for independent operation. Clutches shall have sufficient capacity to give each drum a bare drum line pull of not less than 15,000 pounds, and a maximum speed of not less than 190 feet per minute (fpm). Brakes shall have sufficient capacity to brake and hold a load of not less than 20,000 pounds on the bare drum. The brake lever shall permit weight feeding and shall be provided with a positive locking mechanism. Each drum shall have a spooling capacity of not less than 200 feet of 5/8-inch diameter wire rope and shall be furnished with not less than 160 feet of 5/8-inch diameter, 18 x 7 non-rotating, hemp core, improved plow steel, preformed, uncoated steel, wire rope. The free end of both ropes shall be fitted with an open socket for securing rope ends to the two eye anchors as specified in 3.13.2 or to the 3-part line traveling blocks. Two single sheave, steel, 2-part or 3-part line traveling blocks shall be furnished with the rig. Sheave diameter shall be not less than 10 inches. Hoisting capacity of the traveling blocks shall be not less than 30,000 pounds. The pipe hoisting travel block shall be equipped with a swivel type safety hook. A single weight indicator shall be provided which is compatible with the traveling block furnished and shall have a capacity commensurate to the mast. The weight indicator gage shall be located at the drilling station and shall be equipped with an adjustable target pointer. The weight indicator shall automatically read whichever line is in use.

3.13.3.2 Sand-reel hoist or third drum. The sand-reel hoist clutch shall have sufficient capacity to give a bare drum line pull of not less than 4,000 pounds and a maximum bare drum speed of not less than 300 fpm. Brakes shall have sufficient capacity to brake and hold a minimum load of 5,000 pounds on the bare drum. The drum shall have a spooling capacity of not less than 1,500 feet of 3/8-inch diameter wire rope, and shall be furnished with not less than 1,200 feet of 3/8-inch diameter, 6 x 19, hemp core, improved plow steel, right regular lay, preformed, uncoated steel, wire rope. The free end of the rope shall be fitted with a safety clevis.

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3.13.3.3 Cathead. The cathead shall have a speed range from 30 fpm to in excess of 190 fpm. The barrel of the cathead shall be not less than 8 inches in diameter. The cathead shall be so mounted as to be accessible for use from the ground, and shall be fitted with a safety device to preclude the possibility of rope turns overlapping.

3.13.4 Dust collection system. The drilling machine shall be furnished with a dust control system including, but not necessarily limited to, a diverter, water tank, and water injection pump. The water tank shall have a capacity of not less than 50 gallons. The tank filler cap shall be equipped with a removable strainer that shall prevent the entrance of foreign matter into the tank. The water pump shall have a minimum discharge capacity of 8 gallons per minute (gpm) at 275 pounds per square inch (psi). The pump's discharge line shall connect to the discharge line of the air receiver tank.

3.13.5 Blowdown hose. A 3/8-inch by 50 foot blowdown hose shall be provided to inflate tires and clean the drilling machine. The hose shall be provided with quick disconnect couplings, nozzle, and tire valve chuck. The hose shall be compatible with the pressure of the air source not to exceed 125 psi.

3.13.6 Rotary assembly. The rotary assembly shall include a transmission and rotary table.

3.13.6.1 Rotary transmission. Power to the rotary table shall be transmitted from the engine through the transfer case and a rotary transmission. The rotary transmission shall have not less than five speeds forward and one speed reverse. When the engine is operating at maximum governed speed, the rotary transmission shall provide a maximum speed of not less than 150 revolutions per minute (rpm) to the kelly bar. By gear selection the rotary transmission shall provide a minimum speed of not more than 20 rpm to the kelly bar. Maximum torque output at the kelly bar shall be sufficient to meet the performance requirements specified herein. The transmission shall be fitted with accessible drain and filler plugs. Levers for shifting gears shall be provided at the drilling station.

3.13.6.2 Rotary table. The drilling machine shall be furnished with an enclosed, oil bath lubricated, heavy duty, rotary table having a minimum opening of not less than 7-1/2 inches. The enclosed rotary table shall be fitted with oil drain and filler plugs accessible for servicing. The table shall be equipped with a kelly bar drive bushing. The bevel ring gear and pinion shall be hardened alloy steel with ground, spiral cut teeth. The pinion shaft shall be mounted on heavy duty antifriction type bearings. The table shall be equipped with double acting hydraulic cylinder(s) to allow horizontal movement of the table. Movement of the table from its normal operating position shall be sufficient to allow passage of 14 inch-diameter tools into the hole being drilled. Table movement shall be controlled by the operator at the drilling station.

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3.13.6.3 Power breakout. A power breakout mechanism shall be provided. Breakout tongs shall be furnished to permit breaking of joints with power. The breakout mechanism shall swing clear for the running of large drill collars and castings. The hydraulic breakout cylinder shall be furnished for use with tongs for breaking tight drill pipe connections.

3.13.7 Kelly bar and air/mud swivel. The kelly bar shall be heat-treated steel with a diameter of not less than 4-1/8 inches and shall be made of a length to accommodate the 20-foot drill pipe. The kelly bar shall have not less than three drive flutes. The kelly bar shall be compatible with drive bushings located in the rotary table that apply rotary drilling motion to the drill pipe, leaving the kelly bar free to move vertically. The kelly bar shall be equipped on the lower end with a saver sub for the drill pipe. The kelly air/mud swivel shall be equipped with a bail. The swivel shall have a capacity of not less than 25 tons and shall be furnished with a swivel sub if required for connecting to the kelly bar. The swivel gooseneck shall be compatible with the discharge hose and minimum 2-inch water course.

3.13.8 Pulldown system. Power to operate the pulldown system shall be transmitted from the engine through a hydraulic transmission. The hydraulic transmission shall have sufficient power to provide a maximum pulldown force of not less than 25,000 pounds on the swivel yoke. The pulldown system shall also have the capability of "holding back" weight on the drill string. The pulldown system shall provide continuous feed over the full length of the kelly bar. Controls shall be provided to control feed and retraction of the drill string. The feed control shall operate smoothly and shall provide variable feed rates. Controls shall be located at the control console. The pulldown system shall be provided with two 1-1/2-inch pitch roller chains.

3.13.9 Mud pump. The mud pump shall be the horizontal, duplex, double-acting piston type. The valves and valve seats shall be hardened alloy steel having a minimum hardness of 600 Brinell. The piston shall be a one-piece alloy steel body containing replaceable double lipped cups of abrasive-resistant rubber. The pump shall have cylinder liners of heat-treated alloy steel having a minimum hardness of 600 Brinell. The piston rod shall be corrosion-resistant steel having a minimum hardness of 500 Brinell. Lubricated packing glands shall be provided on the piston rod. The pump shall have a capacity of not less than 180 gallons of mud fluid per minute at a pressure of 400 psi when operating at 70 strokes per minute. Wrenches other than those specified herein and liner and valve seat pullers necessary to perform maintenance on the pump shall be furnished. Provisions shall be made to drain the pump and line in their mounted positions.

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3.13.9.1 Pump drive. The pump shall be driven by the power unit. The pump drive shall assure that the speed of the pump is not affected by changes in speed of the rotary assembly. Power from the power unit shall be transferred to the pump through clutch and reduction drives. All chain and gear drives shall be oil bath lubricated.

3.13.9.2 Discharge line. The discharge line shall consist of piping, air chamber, pressure gage, relief valve, control valves, flexible connection to the mast pipe, and the mast pipe. The line shall connect to the discharge side of the pump and run to a point near the mast with a flexible hose connection to a mast pipe extending one-half the height of the mast. The top end of the mast pipe shall be equipped with a hammer union for connecting to the discharge hose. The discharge line shall include a bypass line with control valve for cutting off and bypassing the drilling fluid. The discharge line shall also include an outlet with control valve for connecting the 1-1/2 inch inside diameter (ID) wire wrapped hose and an outlet with control valve for connecting a 2-inch ID discharge hose from an auxiliary mud pump. The pressure gage shall be a 4-1/2-inch dial type, visible to the operator and calibrated for pressure from 0 to 1,000 psi. The gage shall be protected to prevent any fluids and colloids from entering the gage tubes. The relief valve shall be adjustable for 1,000, 850, 600, 500, 400, and 250 psi settings.

3.13.9.3 Discharge hose. The discharge hose shall be 2 inches ID and adequate length to permit full kelly bar travel. The hose shall be equipped with a hammer union on one end for connection to the top mast of the mast pipe and a fitting on the other end for connection to the kelly swivel. Threads shall be standard National Pipe threads (NPT). A safety chain on both ends of the high pressure discharge hose shall be furnished.

3.13.9.4 Suction line. The suction line shall consist of a steel pipe or fitting, not less than 4 inches in diameter, extending from the pump to an accessible location for attachment of the suction hose. The free end of the pipe or fitting shall be fitted with a quick couple hammer union.

3.13.9.5 Suction hose and strainer. The suction hose shall be not less than 4-inches ID by 20 feet long and shall be equipped with hammer union couplings and a perforated cylindrical strainer. Threads shall be standard NPT.

3.14 Air system. The drilling machine shall be equipped with a type A or B air system as specified (see 6.2) for air/rotary drilling and bottom hole percussion/rotary drilling.

3.14.1 Type A air system. When a type A air system is specified, the air system shall include an air compressor, receiver tank, air line lubricator, and a discharge line as specified herein.

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3.14.1.1 Air compressor. The air compressor shall deliver not less than 550 cubic feet of free air per minute at 40 psi at 950 rpm or 470 cubic feet of free air per minute at 250 psi at 950 rpm when pumping continuously. The compressor shall be supplied with a dual pressure unloading system.

3.14.1.2 Air receiver tank. The air receiver tank shall have a capacity of not less than 7-1/2 cubic feet. The tank shall be constructed in accordance with the ANSI/ASME Boiler and Pressure Vessel Code for Unfired Pressure Vessels.

3.14.1.3 Air discharge line. The air discharge line shall consist of a 0 to 500 psi pressure gage, a shutoff valve and 2-inch piping to mast pipe hose.

3.14.2 Type B air system. When a type B air system is specified, the air system shall include an air receiver manifold and an air line indicator as specified herein.

3.14.2.1 Air receiver manifold. The air receiver manifold shall connect one 600 cubic feet per minute, 100 psi, auxiliary air compressor to the drilling machine. The receiver shall conform to ANSI/ASME Boiler and pressure vessel code, section VIII. The manifold shall be provided with a pressure relief safety valve set at 125 psi, a moisture drain valve, two inlets and one 2-inch ID outlet for connecting the discharge line. The two inlets shall be 2-inch NPT nipples equipped with caps. The discharge line shall consist of piping, one-way valve, pressure gage, relief valve, and control valves. The discharge line shall connect the air receiver manifold to the lower mast pipe hose as specified in 3.13.9.2. The one-way valve shall prevent the entrance of water into the air receiver manifold from the water injection pump. The discharge line shall have two inlets equipped with control valves for connecting the discharge lines of the air line lubricator and water injection pump. The pressure gage shall be dial type calibrated for pressure from 0 to 500 psi, and shall be protected against the entrance of fluids and colloids. The pressure gage shall be visible to the operator at the drilling station. The discharge line pressure relief valve shall be the adjustable type within the range 50 to 500 psi. All threads shall be NPT.

3.14.2.2 Air line lubricator. The air line lubricator shall be positive displacement type having a minimum reservoir capacity of not less than 3 gallons. The discharge rate shall be adjustable. The lubricator discharge hose shall be connected to the air discharge as specified herein. The lubricator shall be mounted on the drilling machine in an accessible location for servicing and shall be equipped with a glass sight gage. A red signal light located at the drilling station shall indicate when the lubricator is operating.

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3.15 Drilling station. The drilling machine shall be equipped with a drilling station containing all the controls and instruments specified herein. The drilling station shall be so located as to assure the operator a clear view of the drill pipe at its point of entry into the well. All controls shall be accessible to the operator in his normal operating position. Instruments and gages shall be visible to the operator in his normal operating position. Each instrument, gage and control shall be provided with permanently affixed identification markings.

3.16 Hydraulic system. The hydraulic system on the drilling machine shall provide operating functions as specified herein. The system shall have a working pressure (relief valve setting) that does not exceed 2,000 psi. Oil for the system shall conform to MIL-L-2104, grade 10. All hydraulic component materials shall be compatible with this oil and shall operate with an oil operating temperature not to exceed 250° F at any ambient temperature up to 125° F. Unless otherwise specified herein, all system components shall withstand a proof pressure that is equal to or greater than 1-1/2 times the maximum working pressure (1-1/2 times relief valve setting) without external leakage, damage, or permanent deformation. Unless otherwise specified herein, all hydraulic components shall have a minimum burst pressure that is equal to or greater than 3 times the system relief valve setting. Manufacturer's certification of proof and burst pressure shall be acceptable. The hydraulic system shall be cleaned free of foreign matter (including weld slag and spatters). Hydraulic connection ports of the hydraulic system components shall be sealed by the component manufacturer and this seal shall be maintained until final assembly on the drilling machine. The pump shall supply a continuous flow of hydraulic fluid, and shall have sufficient capacity to energize all hydraulically powered components. A wiper shall be provided on each hydraulic cylinder adjacent to the piston rod seal to prevent dirt and abrasives from damaging the seal or entering the cylinder. Valves shall control retracting speed of all hydraulic cylinders, or an internal flow restriction or bypass shall be provided at each end of the cylinders. The reservoir shall be baffled and shall be constructed to withstand vibrations and shocks. The reservoir shall have sufficient capacity to allow for maximum displacement of all cylinders at the same time, and to aid the heat dissipation. The return line shall terminate below the minimum oil level of the reservoir. The suction line shall be above the reservoir bottom, and shall be protected with an accessible strainer. The reservoir shall be equipped with a removable cover for inspection and cleaning. Provisions shall be made for filling and draining the reservoir and for checking the quantity of fluid in the reservoir. The drain plug shall be of the permanent magnetic type.

3.16.1 Hydraulic lines. Hydraulic lines shall consist of tubing or pipe, hose and fittings. Hose or a combination of hoses and tubing shall be used between components that are not mounted to nonflexing, rigidly connected structural members or rigidly connected sub-assemblies. All other lines shall be tubing. All lines shall be routed to provide the shortest line, minimum number of bends, ease of maintenance, and maximum protection. Provisions shall be made

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to prevent damage of lines due to chafing. The hose installation, such as angle of fittings and location, shall cause no stress concentration on the hose at the fitting, and the hose shall be supported, when necessary, to eliminate sagging at the fitting. All lines other than suction lines shall be considered pressure lines. The pump suction line (pump inlet line) shall include provision for flexibility when there is vibration or movement between the reservoir and the pump. The vacuum reading at the suction side of the pump shall not exceed 5 inches of mercury. Pressure hose shall conform to SAE J517 and pressure hose couplings shall conform to SAE J516. The hose selected shall have a working pressure which is equal to or greater than the hydraulic system maximum working pressure. The hose installation shall not cause the hose to be bent less than the minimum bend radius specified in SAE J517.

3.16.2 Pressure tubing. Pressure tubing and piping shall have a minimum burst pressure of 3-1/2 times the system maximum working pressure. All tubing bends shall be smooth without flattening, kinking, or wrinkling of the tube. The term "without flattening" is defined as follows: "Throughout the bend, the tube maximum diameter shall be equal to or greater than the tube nominal diameter, and the tube maximum diameter minus the tube minimum diameter shall be not greater than 15 percent of the tube nominal diameter." Unless otherwise specified herein, pressure tube fittings shall be of the flare type or double flare type employing 37 degree angle seats. Pressure tube fittings shall have a minimum burst pressure that is equal to or greater than the 3-1/2 times the maximum system working pressure.

3.16.3 System filter. The system filter element shall be in accordance with MIL-F-52723. The filter shall maintain a full flow condition for the duration of first article endurance testing as specified herein. The filter bypass may open for system oil at a temperature less than 70° F. All return oil shall pass through the filter except relief oil which may return directly to the reservoir. The filter shall be located so that the element may be removed for examination or replacement without spilling or draining oil from the reservoir. Filters located outside the reservoir, except "spin-on" types shall be equipped with a drain plug. At the maximum system oil flow through the filter and an oil temperature of 150° F plus or minus 5° F, the pressure drop across a new filter assembly shall not exceed 10 psi. The filter assembly shall include a bypass valve with a cracking pressure of not less than 20 psi differential and shall provide for full flow at a pressure differential of not more than 35 psi. The filter housing shall have a minimum proof pressure of 100 psi without permanent deformation, damage, or leakage. The filter assembly shall be equipped with a contamination indicator which may be remotely mounted. The indicator shall be clearly readable without the removal of any components. As a minimum, the indicator shall show when the element is clean and when the filter is beginning to bypass, requiring a change of the element.

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3.16.4 System cleanliness. After final assembly and prior to submission for inspection and acceptance, the hydraulic system shall be subjected to one of the following clean-up procedures, using filters conforming to National Fluid Power Association (NFLDP), recommended standard method T3.10.8.8, exhibiting a B10 of 10 or more, and with the filter assembly equipped with a filter condition indicator:

a. On-the-drilling-machine filtration - The clean-up filter may be placed in the filter housing or in a housing plumbed into the hydraulic system. Interconnect the hoses to bypass all hydraulic cylinders. Run the engine at not less than 1,000 rpm, actuating all hydraulic functions, including flow over relief valves, for a total of 20 minutes. Increase engine speed to maximum governed speed, actuating all hydraulic functions including flow over relief valves, for 10 minutes. Decrease engine speed to not less than 1,000 rpm, and actuate all hydraulic functions for an additional 30 minutes. (Total filtration time: 60 minutes.) Stop engine and replace system filter element.

b. Off-board filtration - When an off-the-drilling-machine pump and filter assembly is used for clean-up the following procedure shall be used:

(1) Use a pump with a flow in gpm of not less than 1/3 of the oil volume normally in the reservoir. (Example: for a reservoir with 9 gallons of oil, use a pump rated at 3 gpm or more.)
(2) Connect the pump to the reservoir in order that the oil is drawn from the bottom 1/3 of the reservoir. Hoses may be placed into the reservoir through the fill hole or access plate. Run the clean-up pump for 30 minutes.
(3) Interconnect the hose to bypass all hydraulic cylinders. While the clean-up assembly continues to run, start the engine and run to approximately 1,000 rpm pumping oil through each circuit for a minimum of 5 minutes in each direction for all functions.
(4) Reconnect all lines to the appropriate cylinders.
(5) Run engine at approximately 1,000 rpm, actuating all functions, including flow over relief valves, for a total additional time 10 minutes.
(6) Increase engine speed to maximum governed speed, actuating all functions including flow over relief valves for 10 minutes.
(7) Decrease engine speed to approximately 1,000 rpm, and actuate all functions for an additional amount of time so that total filtration time equals 90 minutes.
(8) Stop engine and replace system filter element and remove clean-up assembly.

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3.17 Illumination. The drilling machine shall be provided with a 24 V lighting system to permit daylight performance to be maintained during night operations. Electric power for the lights shall be obtained from the 24 V electric system of the power unit. The system shall include a duplex outlet located on the power unit, necessary switches, wiring and lighting units. All wiring shall be protected against wearing surfaces and weathering.

3.18 Battery box. A weatherproof steel box not less than 0.0747 inch thick (14 gage) shall be provided for the batteries. Vents shall be provided near the top of the box and drains shall be provided in the bottom. The box shall be equipped with built-in restraints constructed so that pressure will be applied to the outer battery case only. When the batteries are installed in the box and with all electrical connections completed and when the battery box lid is in the closed position, there will be no uninsulated electrical conductor within 1 inch of any point within the battery box at which an electrical potential exists. The box shall be located so that the batteries will be accessible for routine maintenance, removal and replacement. The battery electrolyte level shall be visible when the battery filler caps are removed. The batteries shall be accessible for servicing with battery servicing aids such as a hydrometer and a syringe. Grommets shall be provided to protect the battery lead cables. All interior surfaces of the box shall be treated with acid-resistant paint conforming to TT-C-494, type II.

3.19 Handtools. The following handtools shall be furnished with each drilling machine:

Item	Quantity
1. Brushes:	
(a) Wire, 14-inches long	6
(b) Glue, 1-3/4-inches in diameter	2
2. Complete set of mud pump wrenches	1
3. File, bastard, mill, with handle, 12-inches long	2
4. Hammers:	
(a) Sledge, double faced, 10 pounds	1
(b) Ballpeen, 1-1/2 pounds	2
5. Knife, putty, 1-1/4-inches wide	2
6. Lubrication tools:	
(a) Grease gun, lever, hand operated	1
(b) Handpump oiler	1
(c) Pressure type oil can	1

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Item	Quantity
7. Pick, with handle	1
8. Shovel, round point	2
9. Pipe wrenches:	
(a) 24-inch size	2
(b) 36-inch size	2
10. Wrench, open end, adjustable, 12 inch	2
11. Allen wrenches with assortment of setscrews	1
12. 8-ton hydraulic jack	1
13. Pliers, 6 inch	2
14. Screwdriver set 8- to 12-inch	1
15. Set, commercial and technical manuals (TM)	1

3.20 Accessories for down-the-hole percussion drilling. The following cessories shall be furnished with each drilling machine:

Item	Quantity
1. A53-15, Mission Megadril, 5-3/8-inch OD minimum, bottom-hole percussion tool, with 3-1/2-inch API reg pin	1
2. Bits, 6 inch, with button type tungsten carbide inserts (compatible with A53-15)	2
3. Bits, 6-1/2 inch, with button type tungsten carbide inserts (compatible with A53-15)	4
4. Wrenches:	
(a) Bit breakers for bottom hole percussion tool bits:	
(1) Button type, 6 inch	1
(2) Button type, 6-1/2 inch	1
(b) Tong, tool, chain, noncrushing friction type, for disassembling bottom hole percussion tool	2
5. Breakout fork, Megadril Top Sub	1

3.21 Accessories for air/mud rotary drilling. The following accessories shall be furnished with each drilling machine:

Item	Quantity
1. 3-1/2 inch by 20 foot flush joint drill pipe, 0.368 wall, 2-3/8 inch API Internal Flush (IF) flush joint couplings and breakout flats 1-1/2-inch long.	90
2. Same as 1 except 5 foot lengths	3
3. Sub, saver, compatible with kelly bar and 2-3/8 inch API-IF pin	1

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Item		Quantity
4.	Bailer, sectional, 4-1/2-inch outside diameter (OD) 20 feet long two 10-foot sections threaded with dart valve on bottom and hoisting bail on top	1
5.	Drill collar, 4-1/2 inch by 20 foot with 3-1/2 inch API-Reg box up and pin down connections and breakout flats 1-1/2-inch long	4
6.	Breakout fork for breaking out 3-1/2 inch flush joint drill pipe in rotary table	1
7.	Breakout fork for breaking out 4-1/2 inch drill collar in rotary table	1
8.	Bit, rock roller, 3 cone for drilling both air and/or mud rotary	
	(a) Medium soft formation	
	(1) 6 inch with 3-1/2 inch API-Reg pin	4
	(2) 7-7/8 inch with 4-1/2 inch API-Reg pin	2
	(3) 9-7/8 inch with 6-5/8 inch API-Reg pin	2
	(4) 13-3/4 inch with 6-5/8 inch API-Reg pin	1
	(b) Medium hard formation	
	(1) 6 inch with 3-1/2 inch API-Reg pin	2
	(2) 7-7/8 inch with 4-1/2 inch API-Reg pin	2
	(3) 9-7/8 inch with 6-5/8 inch API-Reg pin	2
	(4) 13-3/4 inch with 6-5/8 inch API-Reg pin	1
	(c) Hard formation	
	(1) 6 inch with 3-1/2 inch API-Reg pin	1
	(2) 7-7/8 inch with 4-1/2 inch API-Reg pin	2
	(3) 9-7/8 inch with 6-5/8 inch API-Reg pin	2
9.	Bit, three wing, three blade reaming stabilizer with four sets replaceable blade for:	
	(a) 6 inch with 3-1/2 inch API-Reg pin	1
	(b) 7-7/8 inch with 4-1/2 inch API-Reg pin	1
	(c) 9-7/8 inch with 6-5/8 inch API-Reg pin	1
	(d) 13-3/8 inch with 6-5/8 inch API-Reg pin	1
10.	Bit breakers compatible with above bits (6, 7-7/8, 9-7/8, and 13-3/4 inch)	4
11.	Subs, rock bit, 3-1/2 inch OD, compatible with breakout fork	
	(a) 2-3/8 inch API-IF box to:	
	(1) 3-1/2 inch API-Reg box	3
	(2) 4-1/2 inch API-Reg box	2
	(3) 6-5/8 inch API-Reg box	2
	(4) 2-3/8 inch API-Reg box	1
	(5) 2-3/8 inch API-IF box	1
	(6) 2 inch NPT box	1
	(7) 2 inch NPT pin	1

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Item	Quantity
12. Sub, rock bit, 4-1/2-inch OD, compatible with breakout fork	
(a) 3-1/2 inch API-Reg box	
(1) 3-1/2 inch API-Reg box	2
(2) 4-1/2 inch API-Reg box	2
(3) 6-5/8 inch API-Reg box	2
13. Subs, adaptor	
(a) 2-3/8 inch API-IF box to 3-1/2 inch API-Reg pin	2
14. Clamps, casing, two piece, with four bolts and nuts for:	
(a) 6.625-inch OD casing	1
(b) 8.625-inch OD casing	1
(c) 10.750-inch OD casing	1
15. Hook, well screen, with 2 inch NPT pin	1
16. Mixer, mud, 1-1/2 cubic foot hopper, 1-1/2 inch NPT swivel type female coupling fluid inlet	1
17. Parts, mud pump:	
(a) Packing slush service, sets	2
(b) Packing oil stop head, set	1
(c) Valve springs	3
(d) Valve cover place gasket	8
(e) Valve inserts	8
(f) Fluid cylinder head gasket	2
(g) Slush-proof piston	2
(h) Piston rod	2
(i) Piston liner	2
(j) Liner packing spacer ring	2
(k) Liner packing	4
18. Plug, hoisting, ball bearing swivel, bail type, with 2-3/8 inch API-IF pin, 25,000 pound capacity	2
19. Swab, valve type, 2-3/8 inch API-Reg pin flexible wire basket protectors for each rubber cup, furnished in sets with two cups per set for:	
(a) 6.625-inch OD casing	2
(b) 8.625-inch OD casing	2
(c) 10.750-inch OD casing	2
20. Swage, casing, reshaping, 2 inch NPT box connection for:	
(a) 6.625-inch OD casing	1
(b) 8.625-inch OD casing	1
(c) 10.750-inch OD casing	1

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Item		Quantity
21.	Tools, fishing, with 2-3/8 inch API-IF box connection, for 3-1/2 inch drill pipe with 3-1/2-inch OD flush tool joints:	
	(a) Overshot, circulating and releasing, slip type, with slips	1
	(b) Overshot, die, female	1
	(c) Tap, taper	1
22.	Wrenches:	
	(a) Tong, collar, nonadjustable, open compatible with drill collar wrench flats	2
	(b) Tong, pipe, chain, adjustable for 2-to 12-inch OD pipe	2
23.	Supplies:	
	(a) Chain, close link, welded, proof coil, 3/8 inch, 10 feet long with direct eye slip hook (for 3/8 inch chain) on each end	1
	(b) Lubricant, tool joint compound, graphite base, 10 pound cans	5
	(c) Mud, drilling, Wyoming Bentonite, 100 pound bag	10
	(d) Strainer, tea, 6-inch diameter, No. 30 mesh screen, with wood handle	3
24.	Drilling mud, high yield type, 200 barrels per ton, 50 pound bags.	100
25.	Slips to be used with the following size casing:	
	6.625-OD casing (set)	1
	8.625-OD casing (set)	1
	10.750-OD casing (set)	1
26.	Elevators, regular pattern type, to fit the following size casing and with the links as indicated:	
	6.625-inch casing, 1-1/4 by 22 inch links	1
	8.625-inch casing, 1-3/4 by 26 inch links	1
	10.750-inch casing, 1-3/4 by 28 inch links	1
27.	Lubricant, tool joint compound, graphite base, 10 pound cans.	5
28.	Pipe wrench, 48-inch size	2
29.	Pick mattock with handle	1
30.	Baroid rig mud laboratory in a stainless steel cabinet. The laboratory shall be Baroid Model 821-01 and shall consist of the following:	1
	1 No. 140-01 Baroid 4-scale mud balance	
	1 No. 201 Marsh Funnel Viscometer	
	1 No. 202 Measuring cup, 1,000 milliliter (mL)	
	1 No. 207 Stop watch, 7-jewel	
	1 No. 208 Stop watch cover	

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Item		Quantity
	1 No. 306 Filter press, wall mounting	
	1 No. 328 Screen, spare	
	1 No. 335-14 Air hose, 12 inch	
	1 No. 336-10 Air hose adapter	
	1 No. 400-01 Baroid sand content set	
	1 No. 500-07 Chloride test instructions	
	1 No. 510 Potassium chromate indicator solution, 2 ounce	
	1 No. 522 Silver nitrate 0.001 inch polyethelene bottle, 8 ounce	
	1 No. 523 Silver nitrate 0.01 inch polyethylene bottle, 8 ounce	
	1 No. 527-13 Distilled water in polyethylene bottle, 8 ounce	
	1 No. 624 Wide range hydrion pH dispenser, pH 1-10	
	1 No. 626 Short range hydrion pH dispenser, pH 10-14	
	1 No. 913 Brush, large, with wooden handle	
	1 No. 914-01 Brush, test tube, small	
	1 No. 930 Cylinder graduated, 25 mL	
	1 No. 931 Cylinder graduated 10 mL	
	1 No. 937-01 Titration dish, polyethylene, 140 mL	
	1 No. 959 Pipetee, serological, 1 mL	
	1 No. 961 Pipette, serological, 10 mL	
	1 No. 964-01 Stirring rod, polyethylene	
	1 No. 978 Baroid interval timer	
	1 No. 979-01 Bariod all-metal thermometer 0-350° F	
	2 No. 987 Baroid filter paper (package of 100)	
	1 No. 335 Dead weight hydrolic unit	
31.	Rope, manila, four strand, 3 inch-circumference 1-inch nominal diameter, conforming to T-R-605, testing not required, 200-feet long	1
32.	Spider base assembly with bushings and slips as required to hold 6.625-, 8.625-and 10.750-inch casing.	1
33.	Fishing tool for water pipe 2.50-inch diameter overshot type with guide shoes to fit 6.625 inch-casing.	1
34.	Vise, pipe, chain conforming to GGG-V-415, type I, class 1. Testing not required.	1
35.	Tap, pipe, 2.50 inch, tapered, conforming to GGG-T-581, type I, class 1	1
36.	Diestock, with replaceable pipe dies, conforming to to GGG-T-581, type III, class 6, size A.	1
37.	Couplings, pipe, 2.50 inch NPT, to joint drop pipe	10

3.22 Well completion kit. When specified (see 6.2), well completion kits shall be furnished. A well completion kit consists of the components consummed or committed to a water well and contains the following:

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Item		Quantity
1.	Basket, cementing, 6.625 inch-OD casing. Butler-Larkin Co. Inc., Div. of Valley Industries Inc. Wellsville, NY fig. No. C-535	1
2.	Cable, electrical, single conductor, copper, Number 10 american wire guage (No. 10 AWG), RHW neoprene insulation, jacket dia .20 inch, length 1250 ft.	1
3.	Cable, electrical, submersible pump, 3 conductor, copper, No. 2 AWG, RHW neoprene insulation, jacket dia 1.10 inch, length 1250 ft.	1
4.	Calcium hypochlorite. 81348 O-C-114	3 lb
5.	Cap, well head 97403-13226E1149	1
6.	Casing, well (line pipe), seamless, 6.625-inch-OD by 6.065-in. ID, API 5L, Grade B, threaded and coupled, NPT thd;	
	21 ft-long.	72
	10 ft-long.	1
	5 ft-long.	1
	3 ft-long.	1
7.	Casing, well (line pipe), seamless, 8.625-inch OD by 7.981 inch ID, API 5L, grade B, threaded and coupled, NPT thd; 21 feet long.	10
8.	Casing, well (line pipe), seamless, 10.750 inch OD by 10.020 inch ID, API 5L, grade B, threaded and coupled, NPT, 21 feet long.	5
9.	Cement, Portland, 94 pound bag, ASTM C150.	8
10.	Clamp, pipe. 97403-13226E1150	1
11.	Compound, drilling, organic polymer (Revert), 25 pound bag. Johnson Div. of UOP Inc., St. Paul, MN	15
12.	Compound, inorganic, granular 2 pound package (fast-break). Johnson Div. of UOP Inc., St. Paul, MN	
13.	Connector, insulated, 2-screw type, 1.5 inch NPT. 81348 W-F-406, type I, class 4, style M	1
14.	Connector, straight, liquid tight, strain relief w/neoprene bushing, NPT, 0.50 inch, cable dia .125 inch minimum and .250 inch maximum. 81348 W-F-408, type I, kind BB, style 2	2

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Item		Quantity
15.	Connector, straight, liquid tight, strain relief w/neoprene bushing, NPT, 1.50 in., cable diameter 1.00 inch minimum and 1.187 inch maximum 81348 W-F-408, type I, kind BB, style 2	2
16.	Conduit, metal, rigid, electrical, elbow, 90°, long radius bend, galvanized steel, .75 size. 81348 WW-C-581, class 1, size .75, type C, style 2	1
17.	Conduit, metal, rigid, electrical, elbow, 90°, long radius bend, galvanized steel, 1.50 size. 81348 WW-C-581, class 1, size 1.50, type C, style 2	2
18.	Centralizer, motor, 5.375 inch-OD, 6.00 inch-ID casing, hinged style. Butler-Larkin Co., Inc. Div. of Valley Industries Inc. Wellsville, NY fig. No. C-852	1
19.	Elevators, pipe, 2.50 inch Drillers Service Inc., Hickory N.C. P/N ES250	1
20.	Flange, steel, 6.00 inch, threaded, drilled and faced. 80204 ANSI B16.5, class 300	1
21.	Fluid, biodegradable, foaming agent, air drilling (quick-foam). NL Baroid/NL Industries, Inc., Houston, TX	30 gal
22.	Formaldehyde, 37 percent solution (formalin). Johnson Div. of UOP, Inc., St. Paul, MN	5 gal
23.	Frame assembly: 97403-13226E1147 - assembly 97403-13226E1148 - skid 97403-13226E1230 - mounting 97403-13226E1231 - rod	1
24.	Fuse, cartidge type, nonrenewable, 600 V, 150 ampere 81348 W-F-1726/10, WF 1726/10-003	3
25.	Gasket, 6.00 in., flat ring type, neoprene, .06 inch thick 80204 ANSI B16.21, class 300	2
26.	Generator Set, diesel engine, 30 Kilowatt, 50/60 Hertz (HZ), model MEP-005A, type I (tactical), class 2 (utility), NSN 6115-00-118-1240.	1
27.	Hose, collapsible, 2 inch by 50 feet long, PVC lined w/male and swivel female ends, 2-11-1/2 American National Hose coupling and fire hose coupling threads (NPSH). 81348 ZZ-H-561, grade A, type I or II, style A, class 2	2
28.	Kit, splicing, cable, electrical, consisting of; Sleeve insulating, waterproof, (for connecting 3, copper conductors, No. 2 AWG, RHW neoprene insulation, jacket diameter 1.10 inch Electro-Products Div. of 3M P/N 8407-16	1

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Item		Quantity
	Insulation Tape, electrical, pressure-sensitive adhesive, plastic, .75 inch wide, black. 81348 HH-I-595-B-108-0	1 Roll
	Insulation Tape, electrical, (rubber, natural & synthetic) .75 inch wide. 81348 HH-I-553-A-1-30	1 Roll
29.	Lubricant, tool joint compound, graphite base, 10 pound can. Aermotor Div. of Valley Industries Inc, Conway, AR	2
30.	Nipple, pipe, close, 2 inch NPT. 96906 MS51953-193	4
31.	Nipple, pipe, close, 2.50 inch NPT. 96906 MS51953-217	3
32.	Nipple, pipe, short, 2.50 inch NPT 96906 MS51953-218	3
33.	Nipple, pipe, 2 inch NPT by 10.00 inches long 96906 MS51953-205	3
34.	Nipple, pipe, 2.50 inch NPT by 12.00 inches long 96906 MS51953-230	4
35.	Nut, plain hexagon, C.S., cadmium plated, .375-16UNC-2B. 96906 MS51967-8	4
36.	Nut, plain hexagon, C.S., cadmium plated, .750-10UNC-2B. 96906 MS51967-23	14
37.	Pipe, 2.50 in., steel, galvanized, threaded and coupled, standard weight, 21 feet long. 81346 ASTM A120	60
38.	Pipe fitting, adapter, straight, pipe to hose, 2 inch male NPT to 2 inch male NPSH. 97403-13218E0479-37	2
39.	Pipe fitting, adapter, pipe to hose, 2 inch male NPT to 2 inch female NPSH. 97403-13218E0479-20	1
40.	Pipe fitting, reducer, bell, galvanized, 2.50 inch to 2 inch NPT. 81348 WW-P-521, type II	1
41.	Pipe fitting, coupling, galvanized, 2 inch NPT 81348 WW-P-521, type II	4
42.	Pipe fitting, coupling, galvanized, 2.50 inch NPT 81348 WW-P-521, type II	3
43.	Pipe fitting, elbow, 90°, galvanized, 2.50 inch NPT 81348 WW-P-521, type II	4
44.	Plastic Sheet, polyolefin, clear, 12 feet by 19 feet by 6 mils thick. 81348 L-P-378, type II, class 1, grade and finish optional	2

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Item		Quantity
45.	Pump, submersible, w/special high pressure construction; capacity 50 gpm at 1200 feet of discharge head; 2-1/2 inch FPT discharge; weight 470 pounds, length 231 inches; bowl diffusers, impellers, motor bracket, and discharge head to be molded from CDA 836 bronze; shaft to be type 416 stainless steel w/hard chrome plate; extra seal to be provided for motor guard; a 3 foot length of submersible electric cable, 3 conductors, copper, No. 2 AWG, RHW neoprene insulation, jacket diameter 1.10 inch, shall be spliced to the three motor leads; splice shall be waterproof. Aermotor Div. of Valley Industries Inc., Conway, AR Model 4HMO, 16 Stage	1
46.	Screw, cap, hexagon head, steel, cadmium plated, .375-16UNC-2A by 1 inch long. 96906 MS90725-60	4
47.	Screw, cap, hexagon head, steel, cadmium plated, .750-10UNC-2A by 4 inches long. 96906 MS90725-195	14
48.	Starter, reduced voltage, autotransformer type, for electric pump motor, NEMA 3R enclosure w/fusible disconnect, ambient compensated adjustable quick-trip overload protectors; 30 horsepower 460 V, 3 phase, 60 Hz; starter contacts must remain engaged with a 50 percent drop in line voltage. General Electric Co. Industrial Sales Div., Hastings, Neb. Part No. CR231-PEE-30HP-460 Volts, Pump Panel with Special Low Voltage Starter	1
49.	Shoe, cement float, 6.625-OD casing. Butler-Larkin Co. Inc. Div. of Valley Industries Inc. Wellsville, NY fig. No. C-503	1
50.	Splice, conductor, copper compression, uninsulated, No. 2 AWG wire, long barrel type. 81348 W-S-610, class 2, style A	6
51.	Strap, tiedown, electrical components, adjustable, self-clinching, plastic, double loop, heavy duty, 15.50 inches long. Electro-Products Div. of 3M P/N 06242	250
52.	Strainer, well water, wire wound, 6.00 inch nominal diameter 10 feet long. 81349 MIL-S-416, M416-6-W-1	2

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Item		Quantity
53.	Tank, fabric, collapsible w/staves, 3000 gallon NSN 5430-00-355-4486	1
54.	Union, pipe, 2.00 inch NPT galvanized. 81348 WW-U-531-18-T-D	1
55.	Union, pipe, 2.50 inch NPT, galvanized. 81348 WW-U-531-19-T-D	1
56.	Valve, check, deep well, 2.50 inch NPT, 350 psi. Techno Corporation, Erie, PA P/N Style 002F	2
57.	Valve, gate, bronze, 125 pound, 2.00 inch NPT. MSS-SP-80, type I, class A, style 1	2
58.	Washer, flat-round, steel, cadmium plated, 3/8-inch nominal size. 96906 MS27183-14	4
59.	Washer, flat-round, steel, cadmium plated, 3/4-inch nominal size. 96906 MS27183-23	28
60.	Washer, lock-spring, helical steel, cadmium plated, 3/8 inch nominal size. 96906 MS35338-46	4

3.23 Semitrailer. The minimum requirements of each semitrailer shall be in accordance with the following:

a. Legal requirements:

(1) DOT Federal Motor Vehicle Safety Standards (FMVSS).
(2) DOT Federal Motor Carrier Safety Regulations (FMCSR).

b. Industry standards:

(1) Society of Automotive Engineers (SAE)
(2) Truck-Trailer Manufacturers Association (TTMA)
(3) Tire and Rim Association (TRA)
(4) American Trucking Associations (ATA)

3.23.1 Construction. The semitrailer shall be constructed to withstand without damage or permanent deformation, the static and dynamic loads imposed by operation and testing as specified in 4.5.2.8.3. The semitrailer shall have oscillating beam, tandem, dual-wheel rear axles equipped with nondirectional cross country (NDCC) standard military tires of suitable size. The semitrailer shall be equipped with waterproof straight airbrakes, upper fifth wheel and king pin assembly; and, when specified (see 6.2) a removable dual wheeled, single axle front dolly. The airbrake system and tracking requirements shall be in accordance with DOT Motor Carrier Safety Regulations Sections 293.40 through

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293.43 and 293.45 through 293.50. The semitrailer airbrake system, upper fifth wheel, electrical system and dolly shall be compatible for towing by military M-Series trucks and truck-tractors. The semitrailer shall be compatible with the M-199 military dolly. The semitrailer shall be provided with two vertical lift, telescopic, nonrotating landing legs, with two speed gears and a handcrank. The handcrank shall be located on the curbside of the semitrailer. When specified (see 6.2), the semitrailer shall be equipped with four, double acting, hydraulic, stabilizing jacks powered by the rig hydraulic system. Extension and retraction of the jacks shall be controlled by the operator at the drilling station. Jacks shall have sufficient capacity to sustain the maximum loading which can be imposed by the drilling machine. Jacks shall be provided with counterbalance valves and mechanical locks to prevent uncontrolled retraction. Each jack shall be equipped with a positive locking device to prevent accidental extension while in a travel position. Two jacks shall be located at the rear corners of the trailer. The cylinders shall have chrome-plated cylinder rods. Two jacks shall be located immediately behind the upper fifth wheel plate in a position that shall conform to the dimensions and clearance as specified in 3.1.1. Mud flaps shall be furnished behind the rear wheels.

3.23.2 Vehicular electrical system. The semitrailer shall be equipped with a waterproof, 24-volt vehicular electrical system for service and night travel. The service electrical system shall be in accordance with the "Electrical Equipment and Lighting" section of the SAE Handbook and the ATA Recommended Equipment Specification E-3. Class A turn lights shall be furnished.

3.24 Dolly. When specified (see 6.2) the drilling machine shall be furnished with a single axle dolly. Rims and tires shall be the same size as those furnished on the semitrailer and of sufficient quantity to support the load. The drawbar shall be hinged to permit vertical adjustment. The hinge pins shall be 36 inches above the ground. The drawbar shall be capable of being locked in any one of eight equally spaced positions so that the lunette ring height above the ground will range from 8 inches to 44 inches. The lunette of the drawbar shall conform to MS51336. The dolly shall be equipped with a fifth wheel which is compatible with the semitrailer. The unladen height of the rigid-type fifth wheel shall be 55 inches plus or minus 1 inch. The overall width of the dolly shall not exceed the overall width of the semitrailer. The dolly shall be capable of turning 90 degrees to either side without interference or damage to the dolly or semitrailer. Two safety chains shall be furnished conforming to the requirements of DOT Motor Carrier Safety Regulations.

3.25 Electromagnetic interference. Unless otherwise specified (see 6.2), the electromagnetic interference emission characteristics of the drilling machine shall conform to MIL-STD-461, class C1.

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3.26 Lubrication.

3.26.1 Lubrication provisions. All surfaces requiring lubrication shall be provided with a means for lubricating. The drilling machine shall operate as specified herein when lubricated with military lubricants (see 6.4).

3.26.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, types I, II and III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and plates equipped with hand-operable, quick-disconnect fasteners may be removed to provide accessibility.

3.26.3 Pressure-release device. A pressure release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.26.4 Grease lubrication. All grease lubrication, including lubrication of sealed bearings, shall be with greases as follows:

a. MIL-G-23827 - For ball, roller, and needle hearings, gears, and on sliding and rolling surfaces of such equipment as instruments and mechanical control systems; for use on gears, actuator screws and other equipment requiring a lubricant with high load-carrying capacity over a temperature range of minus 100° F to plus 250° F.

b. MIL-G-10924 - For lubrication and surface corrosion protection of all automotive equipment operated over the temperature range of minus 65° F to plus 225° F. Also used in other applications within this temperature range where grease with oxidation-resistant and corrosion-preventive properties is desirable. Not for use on machinery which comes in contact with foods or potable water.

The drilling machine shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings should be cleaned before lubrication, because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.26.5 Oil lubrication. The drilling machine shall operate as specified herein when serviced with military oils designated for use in the temperature range specified herein (see 6.4). A tag shall be attached in a conspicuous place to indicate the temperature range, and the lubricant grade used.

3.27 Slinging provisions. Permanently affixed slinging provisions that enable the drilling machine to be lifted to its normal travel position shall be provided and shall conform to MIL-STD-209. Slinging provisions may also be used as tiedowns when such provisions conform to MIL-STD-209.

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3.28 Tiedown provisions. Permanently affixed tiedown provisions that permit fastening the drilling machine to the floor or deck of a transportation medium shall be provided and shall conform to MIL-STD-209. Tiedown provisions may also be used as slinging provisions when such provisions conform to MIL-STD-209.

3.29 Identification marking. The drilling machine shall be identified in accordance with MIL-STD-130. The marking shall be applied to the drilling machine on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A class 1 material. Each plate shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.30 Shipping data plate. A shipping data plate conforming to MIL-P-514, type III, composition C, of type I, grade A, class 1 material shall be provided. In addition, the plate shall show the silhouette of the drilling machine in transport position indicating the center of gravity and the location and capacity of the slinging and tiedown provisions. The plates shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.31 Instruction plates. Each lever, switch, and other type control shall be clearly marked to identify its function and shall include instructions, warnings, and cautions where damage to the machine or danger to personnel could result due to incorrect settings during starting, operation, or shutdown of the drilling machine. Instruction plates shall conform to MIL-P-514, type III, composition C of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.32 Treatment and painting. The portions of the drilling machine normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A.

3.32.1 Stenciling. The gross weight of the drilling machine shall be stenciled on both sides of the drilling machine in such a position as to be legible. The letters shall be gothic capitals and the numerals shall be of similar appearance. The letters and numerals shall be 2 inches high where space permits.

3.33 Transportability.

3.33.1 Air. The drilling machine shall be capable of being transported in a C-130 aircraft and meet the requirements of MIL-A-8421 (except that testing will not be required) and figure 1. Disassembly of the drilling machine is permitted to meet this requirement; however, the degree of disassembly shall be limited to the following:

a. Disassembly must be performed by the tools furnished with the drilling machine (see 3.19).

b. A crane or other lifting device may be used to remove components.

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c. Only that disassembly required to provide clearance while loading and transporting, as specified in MIL-A-8421, for C-130 aircraft shall be performed.

d. Components removed from the drilling machine shall be transported on a separate aircraft.

e. The drilling machine must be loaded aboard the aircraft by backing the semitrailer up the ramp and into the cargo area. If the drilling machine is furnished with a dolly, the dolly will be used to load the drilling machine and remain aboard.

3.33.2 Rail. Unless otherwise specified (see 6.2), the drilling machine shall withstand rail humping without damage or permanent deformation to any part or component when tested as specified in 4.5.2.12. After testing, the drilling machine shall operate as specified herein for a period of not less than 1 hour without malfunction or permanent deformation of any part of the drilling machine.

3.34 Tool storage. When specified (see 6.2), the drilling machine shall be furnished with toolboxes and built-in storage compartments to carry all tools specified herein that are 5 feet and less in length. Boxes and compartments shall be sheet steel. All doors and lids shall be equipped with continuous hinges and latches. Latches shall keep doors and lids securely fastened when subjected to vibrations. In addition to the latches, each box and compartment shall be fitted with a staple, hasp, and padlock. Not less than six keys shall be furnished for the padlocks, and any single key shall open all the padlocks. Toolboxes shall be securely fastened to the rig's bed in accessible locations. Built-in storage compartments shall be located under the rig's bed, but no portion of the compartments shall extend below the lowest extremity of the trailer main frame. Compartment floors shall be decked with plywood. Plywood shall be exterior grade, 5-ply, and not less than 3/4-inch thick. Plywood shall not be spliced together to form a compartment deck. Compartment doors shall open in either a sideward or downward direction.

3.35 Workmanship. The drilling machine shall be free from defects such as incomplete welds, rust, cracks, and other defects that could impair the operation or serviceability of the drilling machine. Maximum use shall be made of interchangeable hardware and fastening devices and of a minimum number of types and sizes of studs, bolts, screws, washers, nuts, and similar common (standard) parts. Except as specified herein, flexible hose, conforming to SAE J1402, type E or F, with matching reusable fittings shall be furnished for engine oil lines, engine fuel lines, and straight airbrake lines. All breathers for lubricant and fluid sumps, brake components, and similar openings on the drilling machine shall prevent the entrance of airborne water and airborne dust.

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3.35.1 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not damage the material. Corners shall be square and true. Flame-cutting using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.35.2 Welding. The surface of parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing and other foreign matter. (Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.) Parent materials, weld filler metals and fabrication techniques shall be as required to enable the drilling machine to conform to the examination and test requirements specified in Section 4. Parts to be joined by fillet welds shall be brought into as close contact as possible and in no event shall be separated by more than 0.1875 inch unless appropriate bridging techniques are used. Unless otherwise specified (see 6.2), the welding process used in fabrication of the drilling machine shall be at the option of the contractor.

3.35.3 Welders. Before assigning any welder to manual welding work covered by this specification, the contractor shall provide the contracting officer with certification that the welder has passed qualification tests as prescribed by either of the following listed codes for the type of welding operations to be performed and that such qualification is effective as defined by the particular code:

ANSI/AWS D1.1, Structural welding code, section 5, qualification.
ANSI/ASME Boiler and pressure vessel code, section IX, welding qualifications.

Contractors who make only horizontal welds need not qualify welders for "all position welding". Subject to approval by the Government, the contractor's standard welder qualification may be substituted in lieu of the above codes provided that the contractor's procedure is equivalent to the above codes. The contractor shall be responsible for determining that automatic welding equipment operators are capable of producing quality welds in accordance with ANSI/ASME or ANSI/AWS codes.

3.35.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement.

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3.35.5 Riveted connections. Rivets shall fill the holes completely. The upset rivet heads shall be full, neatly made, concentric with the rivet holes, and in full contact with the surface of the member, and shall be in accordance with SAE J492.

3.35.6 Machine work. Tolerance and gages for metal fit shall conform to the standards of the well drilling industry.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Inspection comparison (see 4.6).

d. Inspection of packaging (see 4.7).

4.3 First article inspection.

4.3.1 Examination. The first article drilling machine shall be examined as specified in 4.5.1. Presence of one or more defect shall be cause for rejection.

4.3.2 Tests. The first article drilling machine shall be subjected to the tests marked "X" in column 1 of table III. Failure of any test shall be cause for rejection.

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4.4 Quality conformance inspection.

4.4.1 Examination. Each drilling machine shall be examined for the defects specified in 4.5.1. Presence of one or more defect shall be cause for rejection.

4.4.2 Test. Each drilling machine shall be subjected to the test marked "X" in column 2 of table III. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The drilling machine shall be examined as specified herein for the following defects:

101. Components missing or not as specified.
102. Material not as specified.
103. Materials not resistant to corrosion and deterioration or treated to be resistant to corrosion and deterioration for the applicable environment.
104. Dissimilar metals not in accordance with MIL-STD-889.
105. Contractor does not have documentation available for identification of material finishes of treatment.
106. Screw threads not as specified.
107. Weight and dimensions not as specified.
108. Compatibility not as specified.
109. Engine accessories missing or not as specified.
110. Engine control systems not as specified.
111. Deck plates not as specified.
112. Detachable or hinged driller's and helper's platform not furnished.
113. Swivel vise not furnished or not as specified.
114. Deadline anchor and weight indicator not type specified.
115. Mast not as specified.
116. Crown block not as specified.
117. Drawworks hoist not as specified.
118. Spooling capacity of drums not as specified.
119. Line speed not as specified.
120. Wire rope and fittings not as specified.
121. Power breakout mechanism not furnished.
122. Transmission not as specified.
123. Kelly bar and swivel not as specified.
124. Mud pump not as specified.
125. Discharge line not as specified.
126. Discharge hose not as specified.
127. Suction line, hose and strainer not as specified.
128. Control console not as specified.
129. Hydraulic system not as specified.
130. Illumination not as specified.
131. Battery box not as specified.
132. Accessories and tools missing or not as specified.
133. Well completion kit missing or not as specified.
134. Vehicular electrical system not as specified.

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135. Lubrication not as specified.
136. Lubrication fittings not as specified.
137. Pressure-release device missing.
138. Lubrication identification tags missing or not as specified.
139. Identification markings missing, incomplete, or illegible.
140. Shipping data plate not as specified.
141. Instruction plates not as specified.
142. Treatment and painting not as specified.
143. Toolbox and storage compartments not as specified.
144. Workmanship not as specified.

4.5.2 Tests.

4.5.2.1 Test conditions. Prior to test, the drilling machine shall be lubricated with oils and greases as specified in 3.26. Oils shall be those designed for use in the ambient temperature at the place of test.

4.5.2.2 Test schedule.

TABLE III. Test schedule.

First Article	Quality conformance	Test	Test paragraph	Requirement paragraph
1	2	3	4	5
X	-	Engine	4.5.2.8.4	3.11
X	-	Safety	4.5.2.3	3.5
X	-	Human factors	4.5.2.4	3.6
X	-	Noise level measurement	4.5.2.5	3.6.1
X	-	Reliability	4.5.2.6	3.7
X	-	Maintainability	4.5.2.7	3.8
		Performance:		
X	-	Drilling machine	4.5.2.8.1	3.10
X	X	Operation	4.5.2.8.2	3.10
X	-	Mobility	4.5.2.8.3	3.10.1
X	-	Drawworks	4.5.2.8.5	3.13.3
X	-	Compressor unit capacity	4.5.2.8.6	3.14.1
X	-	Electromagnetic interference	4.5.2.9	3.25
X	-	Slinging provisions	4.5.2.10	3.27
X	-	Tiedown provisions	4.5.2.11	3.28
X	-	Rail car impact	4.5.2.12	3.33.2

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4.5.2.3 Safety. Safety evaluation shall be accomplished throughout testing. Nonconformance to 3.5 shall constitute failure of this test.

4.5.2.4 Human factors. The drilling machine shall be evaluated throughout testing to determine the human factor characteristics inaccordance with MIL-STD-1472. Nonconformance to 3.6 shall constitute failure of this test.

4.5.2.5 Sound level measurement.

4.5.2.5.1 Driller's station. Measure the sound level at the driller's station in accordance with MIL-STD-1474 for steady state noise at the operating condition of 3/4 maximum governed engine speed, rotary transmission in third speed gear drilling at a minimum depth of 40 feet while circulating with air. Should the sound level exceed 85 dB(A), an 85 dB(A) contour shall be established to identify the distances and directions from the noise source at which the noise level is equal to 85 dB(A). Nonconformance to 3.6.1 shall constitute failure of this test.

4.5.2.5.2 Exterior spectator. Measure the exterior sound level in accordance with MIL-STD-1474, for exterior noise. Nonconformance to 3.6.1 shall constitute failure of this test.

4.5.2.5.3 Hazard precautions. Measure the noise level at all personnel-occupied areas, such as the crew's station, maintenance, and control positions, while operating at the maximum noise condition for those positions. Should the noise level exceed 85 dB(A) at any position, failure to provide the precautions specified in 3.6.1 shall constitute failure of this test.

4.5.2.6 Reliability. Using the MTBF specified in 3.7, test the drilling machine as specified in 4.6.2.8.1 except as specified herein with "accept" and "reject" criteria in accordance with figure 2. Continue testing until either an "accept" or "reject" decision is reached, testing need not be performed at night; holes shall not be deeper than 200 feet; simulated drilling of a 1,500 foot hole shall be accomplished using the following three techniques: (1) mud rotary, (2) air rotary, and (3) down-the-hole percussion; and the drilling machine shall be moved to start a new hole if the 200 foot depth is reached prior to completion of the test. A failure is defined as any malfunction which causes a cessation of drilling operations, or any malfunction which, if drilling continues, would cause damage to other components and equipment or introduces a personnel hazard; all of which cannot be corrected in a 90 minute shutdown period by two of the crew with the tools and skills at their disposal. Damage or wear to the drill bit shall be excluded from consideration in determining failure of the drilling machine. Replacement of the drill bit during drilling operation shall be considered as part of the operating procedures and shall not be construed as maintenance. Nonconformance to 3.7 shall constitute failure of this test.

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4.5.2.7 Maintainability.

4.5.2.7.1 Maintenance support evaluation. Maintenance operations required during first article testing shall be accomplished to determine conformance to 3.8.1. Failure of the support elements to maintain the drilling machine as specified shall constitute failure of this test. Errors or inadequacies in the manuals shall not be considered in assessing the maintenance ratio or the MTBF.

4.5.2.7.2 Maintenance ratio. The maintenance ratio shall be computed during testing. Nonconformance to 3.8.2 shall constitute failure of this test.

4.5.2.8 Performance.

4.5.2.8.1 Drilling machine. Set up the drilling machine at the drill site. Drill a hole to a minimum depth of 200 feet performing the following operations:

a. Drill at night using the lights mounted on the drilling machine as the only illumination for a period of 2 hours.

b. Set two or more sections of 21-foot-long surface casing having an outside diameter of 8.625 inches.

Gage the hole for minimum hole diameter. Maintain a drilling log and indicate the following:

Formations encountered and at what depth.
Drilling time.
Shutdown and response (including service and maintenance).

Submit to the contracting officer a certified reproducible copy of the drilling log. Inability to complete the test, reduced drilling rate at night, hole diameter less than new bit diameter (loss of gage), drilling log incomplete, or any permanent deformation or breakage of components of the drilling machine shall constitute failure of this test.

4.5.2.8.2 Operation. Set up the drilling machine for hole drilling operation. Start the power plant and perform all the operations required to simulate drilling a hole. All major components and all controls shall be operated to simulate full operation. The test shall also include a complete check of all tools, accessories, and supplies specified in 3.19, 3.20, 3.21, and 3.22. Any malfunction, breakage, or permanent deformation shall constitute failure of this test.

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4.5.2.8.3 Mobility. The drilling machine shall be prepared for on-and off-road movement and shall be coupled either to a military 10-ton 6 x 6 truck-tractor or a commercial 6 x 4 truck-tractor having a minimum rating of 60,000 pounds gross combination weight. Servicing adjustments to the drilling machine, other than scheduled maintenance and prescribed within the maintenance manual, will not permitted during the mobility test. Couple the drilling machine to the truck-tractor and tow as specified in Table IV for a total of not less than 145 miles.

TABLE IV. Road test.

Course	Mileage and speeds
Primary highway	60 miles at variable speeds up to 55 mph.
Unpaved roads	60 miles at variable speeds up to 30 mph.
Cross-country	25 miles at variable speeds up to 10 mph.

During the road test perform the following:

a. For three passes, tow the semitrailer over 7-inch high obstacles such that all combinations of the wheels have passed over the obstacles. Observe tire-to-surface contact, evidence of binding or interference, and, after returning the suspension to normal, measure the diagonals and parallels for evidence of misalinement.

b. For three passes, tow the semitrailer over V-ditches not less than 36 inches wide and 16 inches deep. Negotiate the ditches both perpendicular and at an angle of 45 degrees to the direction of travel.

c. Test the semitrailer tracking for conformance to the tracking requirements of DOT Motor carrier safety regulations.

d. Jackknife the semitrailer at angles up to 90 degrees from straight ahead both to the left and to the right. Inspect for any damage, kinking or stretching of brake hose and electrical harness and for contact or interference between tractor components and trailer chassis.

e. Turn the prime mover towing the semitrailer in a rolling, sharpest-turning circle for two complete turns and repeat in the opposite direction. Examine for interference between tractor and trailer components.

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f. Turn the semitrailer lights on, and immerse the semitrailer in water not less than 30 inches deep for not less than 2 minutes. Stop the semitrailer using the semitrailer brakes and turn off the lights for not less than 10 minutes. Turn on the lights. Tow the semitrailer out of the water and make a brake stop using the semitrailer brakes and observe semitrailer braking for water fade, grabbing, and chatter. Observe all brake system components and all electrical system components, except hermetically sealed lights, and inspect at least two wheel hubs and bearings for leakage or damage.

g. Tow the semitrailer over a paved level test course and, when the semitrailer reaches a speed of not less than 20 mph, signal the driver to make a panic brake stop using both prime mover and semitrailer brakes. Perform the test four times, two times in each direction, allowing not more than 3 minutes between tests. Observe the brake system for brake application and leaks.

h. Tow the semitrailer over a paved level test course and, when the semitrailer reaches 20 mph, cause the semitrailer to go into "emergency" simulating a trailer breakaway by uncoupling the air hoses from the prime mover. While the hoses are uncoupled, observe the brake system for brake application and leaks. Recouple the brake hoses and test brake coupling for leaks. Disconnect the semitrailer emergency line hose and observe if there is an air leak at the semitrailer coupling.

i. With the semitrailer stopped on a paved surface, uncouple the truck-tractor brake lines from the semitrailer and bleed down all semitrailer air reservoirs. Determine by the drawbar pull method or on actual grades the maximum holding ability of the semitrailer brakes. Release the emergency parking brakes using only tools supplied with each semitrailer or truck-tractor and without resupply of compressed air.

j. Test the semitrailer airbrake system for conformance to the TTMA recommended practice RP No. 12 and for requirements specified in 3.23.

Failure to successfully complete the test shall constitute failure of this test.

4.5.2.8.4 Engine. Prior to assembling the engine to the drilling machine, the engine shall be tested in accordance with the production engine tests and production control tests of MIL-STD-1410. Nonconformance to the requirements specified herein shall constitute failure of this test.

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4.5.2.8.5 Drawworks. Using a weight equal to the total weight of the drill string at a depth of 1,500 feet, suspended from the bottom of the kelly bar, raise and lower the kelly bar through a vertical distance of 2 feet for 50 cycles. Failure to raise the weight and any damage or permanent deformation shall constitute failure of this test.

4.5.2.8.6 Compressor unit capacity.

a. Preparation:

(1) Assemble unit to test stand.
(2) Fill crankcase to proper level.
(3) Connect compressor to air receiver.
(4) Install air cleaner with thermometers.
(5) Install high pressure orifice (0.566 ±0.001 inch diameter).
(6) Open valve on receiver discharge.

b. Procedure:

(1) Set up unit for single stage operation.
(2) Start unit and check for adequate oil pressure. Perform 1-1/2 hour running test at 50 psi and 870 rpm. Adjust pressure using the valve on the receiver discharge. The unloader pilot must be adjusted to unload slightly above 50 psi.
(3) Check for oil and air leakage, excessive vibrations, and overheating; if any are noted, stop the compressor and remedy.
(4) Oil pressure should be 12 psi minimum (hot).
(5) With unit shut down and hot, tighten all unloader line tube fittings and retorque heads and valves.
(6) Reset unloader pilot to unload at 50 psi and load at 40 psi unless otherwise specified. Cycle unit for 5 minutes to check unloader operation.
(7) Convert unit to dual stage operation.
(8) Continue testing as (1) through (6) above, beginning with the 1-1/2 hour run at 150 psi at 870 rpm.
(9) Operate the compressor for 90 minutes at 150 psi at 870 rpm.
(10) Adjust the unloader as required and operate the compressor for 2 hours at 250 psi at 870 rpm.

c. Failure criteria:

Inability to operate the air compressor for 2 hours at 250 psi at 870 rpm or oil pressure below 12 psi shall constitute failure of this test.

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4.5.2.9 Electromagnetic interference. The drilling machine shall be tested as specified in MIL-STD-461. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.5.2.10 Slinging provisions. The drilling machine shall be tested as specified in MIL-STD-209, type II equipment. Any permanent deformation or breakage of the lifting attachments or any other part of the drilling machine or interference between slings and exterior parts of the drilling machine shall constitute failure of this test.

4.5.2.11 Tiedown provisions. The drilling machine anchored by tiedowns other than the eye attachment being tested, shall be tested as specified in MIL-STD-209, type II equipment. Weld failure or permanent deformation of the drilling machine or any component thereon or of any tiedown attachment shall constitute failure of this test.

4.5.2.12 Rail car impact. Anchor the drilling machine on a railroad flatcar, using only the tiedown attachments. Blocking shall be used. Impact the car, with the drilling machine tied down on it, at a speed of not less than 7 mph against a string of five empty cars with the draft gear extended and the brakes set. The semitrailer shall face in the direction of the five cars. Repeat the procedure with the rear of the semitrailer in the direction of the five cars. Evidence of damage or permanent deformation to the drilling machine or any component or part, or movement or loosening of any component or part such as machinery or piping shall constitute failure of this test.

4.6 Inspection comparison. The Government may select drilling machines at any time during the contract production period and subject these drilling machines to the examination and the tests specified in 4.3 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on drilling machines selected at random from those which have been accepted by the Government and will not include the previously inspected first article drilling machine. In addition to any test specified as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.6.1 Inspection failure. Failure of an inspection comparison drilling machine to meet any requirement specified herein during and as a result of the examination and tests specified in 4.6 shall be cause for rejection of the inspection comparison drilling machine and shall be cause for refusal by the Government to continue acceptance of production drilling machines until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on drilling machines

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previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all drilling machines accepted prior to the completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.7 Inspection of packaging.

4.7.1 Preproduction pack inspection.

4.7.1.1 Examination. Examine the preproduction pack for the defects specified in 4.7.2.3. Presence of one or more defects shall be cause for rejection.

4.7.2 Quality conformance inspection of pack.

4.7.2.1 Unit of product. For the purpose of inspection, a completed pack prepared for shipment shall be considered a unit of product.

4.7.2.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.7.2.3 Examination. Samples selected in accordance with 4.7.2.2 shall be examined for the following defects. AQL shall be 2.5 percent defective.

145.	Materials, methods, and containers not as specified for level A or B. Each incorrect material, method, or container shall be considered one defect.
146.	Components for which no specific packaging requirements are specified not preserved in accordance with the general requirements specified for level A.
147.	Disassembly not as specified for level A.
148.	Matchmarking information not applied on the tags with waterproof ink for level A.
149.	Surface requiring a contact preservative in accordance with the referenced document not coated for level A or level B.
150.	Exposed machined surfaces not coated and wrapped or covered as specified for level A or level B.
151.	Semitrailer not preserved in accordance with the referenced document for level A or level B.
152.	Engine and engine accessories not preserved in accordance with the referenced document for level A or level B, or commercial.
153.	Clutches and brakes not preserved as specified for level A.
154.	Gears and gear housings not preserved as specified for level A or level B.
155.	Rotary kelly bar, slip nut, and chuck not preserved and covered or wrapped as specified for level A or level B.
156.	Hydraulic system not preserved as specified for level A or level B.

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157. Drive chains not preserved as specified for level A or level B.
158. Wire rope not preserved as specified for level A.
159. Mud pump not preserved as specified for level A or level B.
160. Flexible hose not coiled and secured and threads protected as specified for level A or level B.
161. Air compressor and associated components not preserved in accordance with the referenced document for level A or level B.
162. Special wrenches, liner and valve seat pullers and hand tools not preserved in accordance with the referenced document for level A or level B.
163. Floodlights, if removed, not preserved as specified for level A or level B.
164. Exposed ends of wiring outlets, and plug openings not sealed with tape for level A.
165. Accessories and supplies not preserved as specified for level A or level B.
166. Components of the well completion kit not preserved in accordance with the referenced document or as specified for level A or level B.
167. Consolidation not as specified for level A or level B.
168. Packing not as specified for level A.
169. Marking missing, illegible, incorrect, or incomplete for level A, level B, or commercial.

5. PACKAGING

5.1 Preproduction pack. Unless otherwise specified (see 6.2), the contractor shall furnish a preproduction pack for examination within the time frame specified (see 6.2) to prove, prior to starting production packaging that the applied preservation, packing and marking comply with the packaging requirements of this specification. Examination shall be as specified in Section 4 and shall be subject to surveillance and approval by the Government (see 6.5). The preproduction pack may be accomplished utilizing either the first article drilling machine or a production drilling machine. If the first article drilling machine is utilized, any preservation and packing shall be removed by the contractor at no expense to the Government when requested by the Government to facilitate comparison between the first article drilling machine and production drilling machines.

5.2 Preservation. Preservation shall be level A, level B, or commercial as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 General requirements. Each complete well drilling machine with all furnished equipment and supplies shall be provided adequate protection to prevent deterioration and damage against the most severe conditions known or anticipated to be encountered during shipment and indeterminate storage. Components for which no specific packaging requirements are contained herein (see 5.2.1.2) and are procured by a Military or Federal specification shall be preserved in accordance with the level A requirements of the applicable document. Packaging of other components for which no specific packaging requirements are specified herein (see 5.2.1.2) shall be determined by the contractor using components of similar design and construction and the criteria established in MIL-P-116 as guidance.

5.2.1.2 Specific requirements.

5.2.1.2.1 Disassembly. Disassembly shall be the minimum necessary to accomplish the transportability requirements specified in 3.33 and to safeguard parts vulnerable to damage, pilferage, and loss, and shall be in accordance with the contractor's recommendations. Bolts, nuts, pins, and washers removed shall be placed in mating parts and secured to prevent their loss. Keys removed shall be secured in the keyway of the pulley or flywheel.

5.2.1.2.2 Matchmarking. When necessary to facilitate reassembly, detached components shall have matchmarking information on cloth tags conforming to UU-T-81, type A. The printing on the tags shall be applied with waterproof ink. The tags shall be securely attached to the components so as to prevent damage to the preservation or to the components.

5.2.1.2.3 Preservatives. Preservatives specified herein shall conform to the applicable specifications listed in and shall be applied in accordance with MIL-P-116. When type P-10 preservative is specified herein, it shall conform to MIL-L-21260, type I, grade 10.

5.2.1.2.4 Unprotected surfaces. Unpainted surfaces of items such as tie rods, adjusting rods, clevises, lubrication fittings, spider base and slips, sprockets, drums, sheaves, bolts, nuts, pins, washers, and screws, or any other unpainted exterior metal surfaces not specified herein requiring the application of a contact preservative in accordance with MIL-P-116, shall be coated with type P-1 preservative.

5.2.1.2.5 Machined surfaces. Exposed machined surfaces of parts shall be coated with type P-6 or P-11 preservative and wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured in place with tape conforming to MIL-T-22085, type II.

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5.2.1.2.6 Semitrailer. The components of the semitrailer chassis and body shall be preserved and packaged in accordance with the level A (mobile) requirements of MIL-STD-281.

5.2.1.2.7 Engine. The engine and engine accessories shall be preserved in accordance with the level A requirements of MIL-E-10062 as specified for type I classification, except the engine cooling system shall be preserved by the water and anti-freeze procedure using anti-freeze conforming to MIL-A-46153.

5.2.1.2.8 Clutches.

5.2.1.2.8.1 Disk type. With the cover plates removed and with the clutch engaged, all interior components of the clutch shall be sprayed with a thin film of primer conforming to TT-P-664 or primer conforming to MIL-P-46093. Coating of nometallic surfaces shall be kept to a minimum. Clutch control mechanisms not enclosed shall be coated with type P-1 preservative. After spraying, spring-loaded type clutches shall be blocked in a partially disengaged position to eliminate contact between the disk facing and pressure plates. Snap-over-center and toggle-in type clutches shall be completely disengaged.

5.2.1.2.8.2 Band type clutches and brakes. Clutch and brake drum facings and control mechanisms enclosed in the housings shall be coated with a thin film of primer conforming to TT-P-664 or primer conforming to MIL-P-46093. Control mechanisms not enclosed shall be coated with type P-1 preservative. The clutch and brakes shall not be operated after application of preservatives.

5.2.1.2.9 Gears.

5.2.1.2.9.1 Enclosed gears. The gear housing shall be filled to the operating level with the lubricant required for operation. The gears shall be actuated to assure complete coverage.

5.2.1.2.9.2 Exposed gears. Exposed gears shall be coated with type P-1 preservative or with primer conforming to TT-P-664 or MIL-P-46093 as required.

5.2.1.2.10 Rotary kelly bar, slip nut, and chuck. The interior and exterior surfaces of the kelly bar shall be coated with type P-10 preservative, and the slip nut and chuck coated with type P-1 preservative. The open ends and the exposed surfaces of the rod shall be covered or wrapped with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured in place with tape conforming to MIL-T-22085, type II.

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5.2.1.2.11 Hydraulic system. The hydraulic fluid supply tank shall be filled with the hydraulic fluid recommended for operation. The pistons shall be fully retracted into the cylinder and secured. Any exposed portions of the piston rods (ramshafts) shall be coated with type P-6 or P-11 preservative and wrapped or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, extending the wrap approximately 2 inches on the ram cylinder. The wrap shall be secured in place with tape conforming to MIL-T-22085, type II. The hydraulic control valves shall be secured in the neutral position and preserved as specified for piston rods (ramshafts). Hose shall not be disconnected unless required to comply with 3.33. Any other exterior metal surface of component parts of the system requiring the application of a contact preservative shall be coated with type P-1 preservative.

5.2.1.2.12 Hoisting assembly. The hoists and the planetary reduction gears shall be coated with type P-1 preservative.

5.2.1.2.13 Drive chains.

5.2.1.2.13.1 Exposed drive chains. Exposed drive chains shall be coated with enough type P-3 or P-9 preservative to assure penetration of the preservative to the inner surfaces of the rollers, pins, and bushings. After the excess preservative has drained, the chains shall be coated with type P-1 preservative. Sprockets shall be coated with type P-1 preservative.

5.2.1.2.13.2 Enclosed drive chains. Enclosed drive chains shall be preserved as specified in 5.2.1.9.1 for enclosed gears.

5.2.1.2.14 Wire rope. The wire rope shall be unwound from the drums and coated with type P-1 preservative. The preservative shall be applied by brushing or by immersion. The drums of the hoists shall be coated with type P-1 preservative and the wire rope rewound on the drum and secured to prevent unwinding.

5.2.1.2.15 Mast. Unpainted metal surfaces of the mast shall be coated with type P-1 preservative.

5.2.1.2.16 Crown block. Unpainted surfaces of the crown block assembly including sheaves, anchor, guard and traveling block shall be coated with type P-1 preservative.

5.2.1.2.17 Mud pump. The fluid cylinders and valve chambers shall be drained. The interior surfaces of the cylinder liners and valve seats shall be coated with type P-10 preservative without disassembly. The exposed piston rod shall be covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured in place with tape conforming to MIL-T-22085, type II.

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5.2.1.2.17.1 Flexible hose. Flexible hose shall be coiled to the minimum size and secured with not less than three equally spaced ties of cotton tape or twine. The threads shall be protected with plastic or composition thread protectors.

5.2.1.2.17.2 Hose fitting and strainers. Hose fittings and strainers shall be preserved in accordance with MIL-P-116, method III.

5.2.1.2.17.3 Special wrenches, liner and valve seat pullers. The special wrenches, liner and valve seat pullers, required for the mud pump, shall be preserved in accordance with the level A requirements of PPP-P-40, as specified for items of like description.

5.2.1.2.18 Air compressor. The air compressor and associated components shall be preserved in accordance with the level A requirements of MIL-C-3600.

5.2.1.2.19 Floodlights. Floodlights, if removed, shall be preserved individually in close-fitting boxes conforming to PPP-B-636, W6c or W5c, style optional. Cushioning to prevent movement and damage shall be provided. Exposed ends of wiring outlets and plugs and plug openings shall be sealed with tape conforming to MIL-T-22085, type II.

5.2.1.2.20 Hand tools. Hand tools shall be preserved in accordance with the level A requirements of PPP-P-40, and placed in the tool box.

5.2.1.2.21 Commercial and technical manuals. Commercial and technical manuals shall be preserved in accordance with MIL-P-116, method IC-1 or IC-3, and placed in the tool box.

5.2.1.2.22 Accessories for down-the-hole percussion drilling and for air/mud rotary drilling. Exterior unpainted and threaded surfaces of accessories such as, but not limited to, bailers, bits, clamps, hook, overshots, and drill pipe, shall be coated with type P-1 preservative. Metal thread protectors shall be used to protect external threads. The interior surfaces of items such as the bailer, and drill bits shall be coated with type P-10 preservative. Openings into the interior of all tools shall be sealed with metal caps or plugs or covered with barrier material conforming to MIL-B-121, type I, grade A, class 2, secured in place with tape conforming to MIL-T-22085, type II. Small accessories requiring the protection afforded by a fiberboard box to prevent mechanical or physical damage shall be placed in a close-fitting box conforming to PPP-B-636, W6c or W5c, as applicable to the weight of the items. The following accessories shall be preserved as specified.

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5.2.1.2.22.1 Mud pump parts. The mud pump parts shall be individually preserved in accordance with MIL-P-116, method IC-1, using type P-10 preservative on the unpainted metal surfaces. The parts shall then be combined together; cushioned to prevent movement or damage and placed in a minimum number of boxes conforming to PPP-B-636, W6c or W5c.

5.2.1.2.22.2 Mud. The drilling mud shall be placed in the specified quantity in sacks conforming to UU-S-48, type II, construction 14-14X.

5.2.1.2.22.3 Chain. Unpainted surfaces of the chain shall be coated with type P-1 preservative.

5.2.1.2.22.4 Brush. Each brush shall be preserved in accordance with MIL-P-116, method IC-1.

5.2.1.2.22.5 Strainer. The strainer shall be placed in a close-fitting box conforming to PPP-B-636, W6c style optional.

5.2.1.2.22.5 Baroid rig mud laboratory. The components of the Baroid rig mud laboratory shall be cushioned, secured, and placed within the cabinet in a manner to prevent movement or damage to the components or cabinet. The cabinet shall then be placed in a box conforming to PPP-B-636, type CF, class weather-resistant, and protected from damage by the use of fiberboard supports or forms within the box. The box shall be closed and sealed as specified for method V in the appendix to the box specification.

5.2.1.2.22.7 Rope. Rope shall be preserved in accordance with the level A requirements of MIL-C-3131.

5.2.1.2.23 Well completion kit.

5.2.1.2.23.1 Generator set. The generator set shall be preserved and packed in accordance with the level A requirements of MIL-G-28554 except the cooling system shall be preserved by the water and antifreeze procedure specified in MIL-E-10062 and then drained. The coolant may be reused in preservation of other systems until it shows evidence of contamination.

5.2.1.2.23.2 Electric cable and wire. Electric cable and wire shall be preserved and packaged in accordance with the level A requirements of MIL-C-12000.

5.2.1.2.23.3 Cement. The cement in the quantity specified shall be placed in sacks conforming to the level A requirements of UU-S-48 as specified therein for portland cement.

5.2.1.2.23.4 Pipe fittings. Pipe fitting shall be preserved and packaged in accordance with the level A requirements of MIL-V-3.

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5.2.1.2.23.5 Electrical components. Electrical components shall be preserved in accordance with the level A requirements of MIL-E-17555.

5.2.1.2.24 Consolidation. Components of the well drilling machine, not including the well completion kit, that can not be placed in tool boxes, and require the protection of a wood box to prevent damage, pilferage or loss or aid in accomplishing the packing specified in 5.3 shall be consolidated together in a minimum number of close-fitting boxes conforming to PPP-B-601, overseas type, style I. Strapping of the boxes shall not be required. The boxes shall be a size and configuration most convenient to fit into the available space on the semitrailer and should be placed and secured in a manner so as not to interfere with lifting or towing. The contents shall be blocked, braced, or anchored as applicable within the boxes to prevent movement or damage. When well completion equipment is specified, securing of those components to the semitrailer is not anticipated, however, those components requiring the protection afforded by a wood box, or to aid in accomplishing the packing specified in 5.3, shall be consolidated in boxes specified herein, except the style shall be optional and the box configuration shall be governed by the packing requirements.

5.2.2 Level B. The complete drilling machine shall be preserved as specified for level A with the following exceptions.

5.2.2.1 Clutches and brakes. Coating of clutch and brake components with primer shall not be required.

5.2.2.2 Hydraulic system. Coating and wrapping of piston rods shall not be required.

5.2.2.3 Exposed drive chains. Exposed drive chains shall be preserved as specified for level A, except only type P-3 or P-9 preservative shall be applied.

5.2.2.4 Wire rope. Coating of wire rope shall not be required.

5.2.2.5 Air compressor. The air compressor and associated components shall be preserved in accordance with the level B requirements of MIL-C-3600.

5.2.2.6 Lighting system. Taping of exposed ends of wiring, outlets, and plug openings shall not be required.

5.2.2.7 Generator set. The generator set shall be preserved in accordance with the level B requirements of MIL-G-28554 except the cooling system shall be preserved as specified in 5.2.3.1.

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5.2.2.8 Consolidation. Consolidation shall be as specified for level A except wood boxes shall be domestic type.

5.2.3 Commercial. Each complete drilling machine with all accessories except for the engines (see 5.2.3.1) shall be preserved in accordance with MIL-STD-1188.

5.2.3.1 Engine. The coolant system, fuel system, and combustion shambers and valves shall be preserved in accordance with MIL-E-10062, as specified for type I classification. Engine cooling system shall be preserved by the water and antifreeze procedure using antifreeze conforming to MIL-A-46153.

5.3 Packing. Packing shall be level A or commercial as specified (see 6.2).

5.3.1 Level A.

5.3.1.1 Surface shipment.

5.3.1.1.1 Without well completion equipment. Each complete semitrailer mounted drilling machine shall be prepared for mobile shipment. Components shall be secured to the semitrailer or the drilling machine in a manner so as not to interfere with lifting or towing and in a manner to prevent movement, pilferage or damage. Securing shall be accomplished by the use of sufficiently sized bolts and nuts; strapping conforming to QQ-S-781, class 1, type I or IV, finish B; or soft annealed wire, as applicable.

5.3.1.1.2 With well completion equipment. Packing shall be specified in 5.3.1.1.1 and the following additional requirements.

5.3.1.1.2.1 Well casing and pipe. Threaded surfaces shall be protected by the use of metal thread protectors. The well casing and pipe shall be shipped loose or bundled into secured lifts as recommended by the contractor. When bundled, bundles shall contain only like items and shall be secured with not less than three equally spaced straps conforming to QQ-S-781, class 1, type I or IV, finish B, size as applicable.

5.3.1.1.2.2 Generator set. The generator set shall be packed in accordance with the level A requirements of MIL-G-28554.

5.3.1.1.2.3 Other components. Components of the well completion equipment, other than the well casing and pipe, and the generator set, whether consolidated or not, shall be packed together in a minimum number of crates conforming to MIL-C-104, type I, class 1 or 2, style a. The contents of each crate shall be securely blocked, braced, and anchored to the crate base to prevent movement or damage in accordance with the appendix to the crate specification and in accordance with MIL-STD-1186.

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5.3.1.2 Air shipment.

5.3.1.2.1 Without well completion equipment. Each complete semitrailer mounted drilling machine shall be prepared for shipment as specified in 5.3.1.1.1, except components and assemblies removed to comply with the air transportability requirements specified in 3.33, and that cannot be relocated and secured to the basic unit, shall be packed in boxes as specified in 5.2.1.2.24 or in crates as specified in 5.3.1.1.2.3. The dimensions of each box or crate shall be of a size compatible for use with the 463L air cargo handling system, using the HCU-6/E pallet as specified in MIL-A-8421. The gross weight of any one crate or pallet shall not exceed 7 tons.

5.3.1.2.2 With well completion equipment. Packing shall be as specified in 5.3.1.2.1 and the following additional requirements.

5.3.1.2.2.1 Well casing and pipe. Well casing and pipe shall be packed as specified in 5.3.1.1.2.1 and in addition, single pieces shipped loose or bundles as applicable shall have snug-fitting wood caps secured over each end. End caps shall be fabricated as shown in the lower portion of figure 2 of MIL-STD-163, except that one end of the box shall be deleted for insertion of the well casing or drill pipe. The caps shall be a minimum of 18 inches in length and shall be secured to each other by strapping conforming to QQ-S-781, class 1, type I or II, finish B, size as applicable. One strap shall be applied over the top and under the bottom of the individual piece or bundle and around the end of each cap at the vertical centerline. A second strap shall be applied around the sides of the individual piece or bundle and over each end cap at the horizontal centerline. Straps shall be secured to the end caps with staples.

5.3.1.2.2.2 Other components. Packing shall be as specified in 5.3.1.1.2.3 except the crate dimensions shall be of a size compatible for use with the 463L air cargo handling system, using the HCU-6/E pallet as specified in MIL-A-8421.

5.3.2 Commercial. Each complete semitrailer mounted drilling machine shall be prepared for mobile shipment in accordance with carrier rules and regulations. When well completion equipment is furnished the components shall be packed in accordance with MIL-STD-1188.

5.4 Marking.

5.4.1 Military. Marking for military levels of protection (level A or B) shall be in accordance with MIL-STD-129. In addition to marking specified, shipping containers and mobile drilling machines shall be marked with not less than 2 inches high letters to indicate "FOR SURFACE SHIPMENT ONLY" or "FOR AIR SHIPMENT ONLY" as applicable.

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5.4.2 Commercial. Marking for commercial packaging shall be in accordance with MIL-STD-1188.

6. NOTES

6.1 Intended use. The drilling machine is intended for use in drilling for water.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. When a first article is required for inspection and approval and the number of units required (see 3.2).

c. Whether a type A or type B air system is required (see 3.14).

d. When well completion kit will be furnished (see 3.22).

e. When a dolly is required (see 3.23.1 and 3.24).

f. When four double acting, hydraulic, stabilizing jacks powered by the rig hydraulic system are required (see 3.23.1).

g. When the electromagnetic interference suppression shall be other than specified (see 3.25).

h. When rail hump test is not required (see 3.33.2).

i. When toolboxes and built-in storage compartments are required (see 3.34).

j. When welding process is not at the contractor's option (see 3.35.2).

k. Time frame required for submission of the preproduction pack when applicable (see 5.1).

l. Degree of preservation and degree of packing required (see 5.2 and 5.3).

6.3 First article. When a first article inspection is required, the item will be tested and should be a first produced item. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, test, and approval of the first article.

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6.4 Lubricants. The contracting officer should furnish a list of military lubricants applicable to the drilling machine covered by this specification as contained in the Federal Supply Catalog, Department of Defense Section, Identification List C9100-IL, for FSC Group 91.

6.5 Preproduction pack. Any changes or deviations of production packs from the approved preproduction pack will be subject to the approval of the contracting officer. Approval of the preproduction pack will not relieve the contractor of his obligation to preserve, pack, and mark the drilling machines in accordance with this specification.

6.6 Drilling requirements. The contracting officer should include provisioning requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operational manuals to be furnished with each drilling machine.

6.7 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions regarding shipment of the drilling machines.

6.8 Noise limits. When the noise level of the drilling machine exceeds 85 dB(A), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474, (MANUALS Paragraph) shall include the requirement for hearing protection, the noise level of the drilling machine, the distance from the drilling machine at which 85 dB(A) will always be met, and what operational situation or system configuration will reduce the level to 85 dB(A).

6.9 Recycled material. It is encouraged that recycled material be used, when practical, as long as it meets the requirements of this specification (see 3.3).

Custodians:
Army - ME
Navy - YD

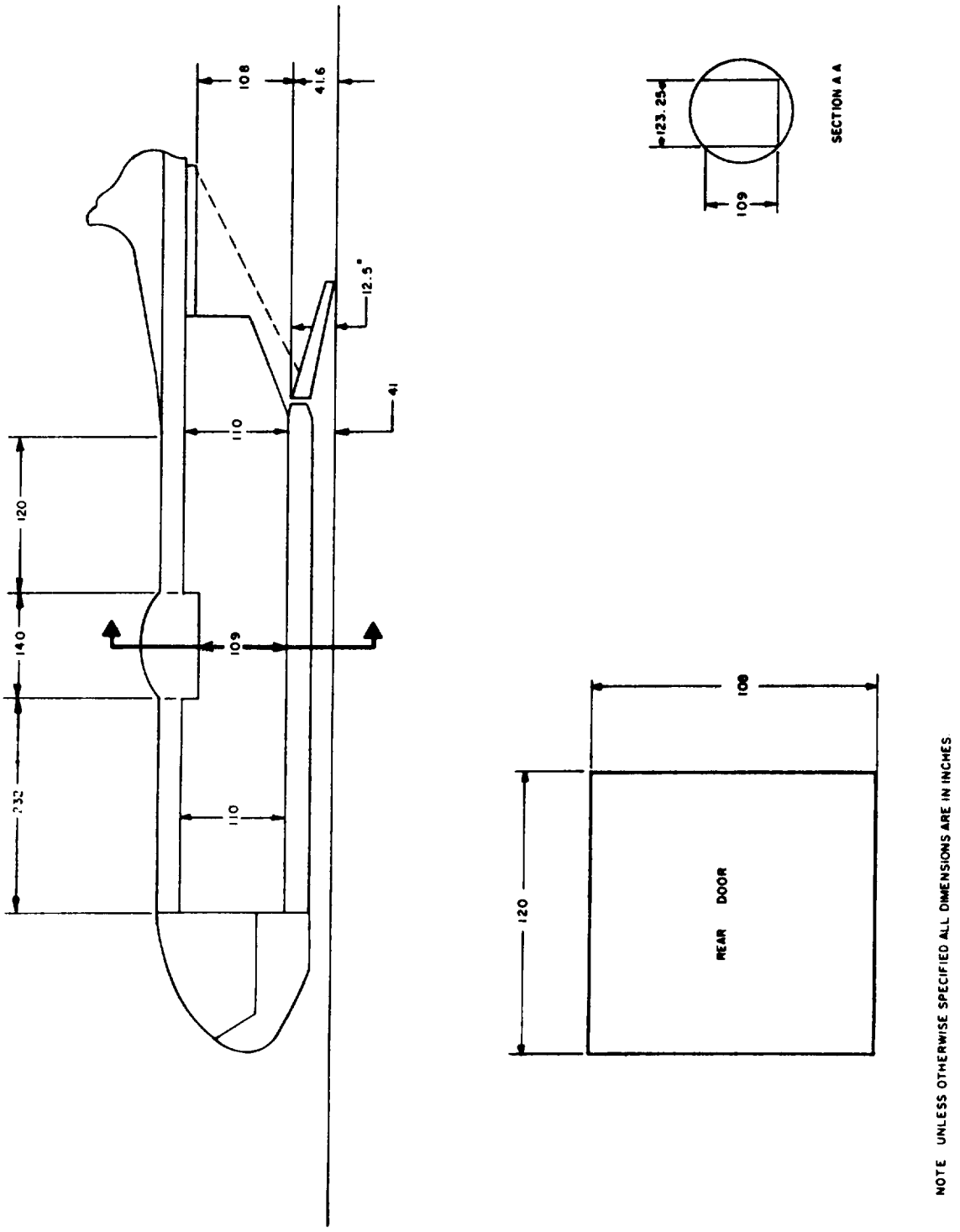
Review activity:
Army - AT

User activity:
Navy - MC

Preparing activity:
Army - ME

Project 3820-0175

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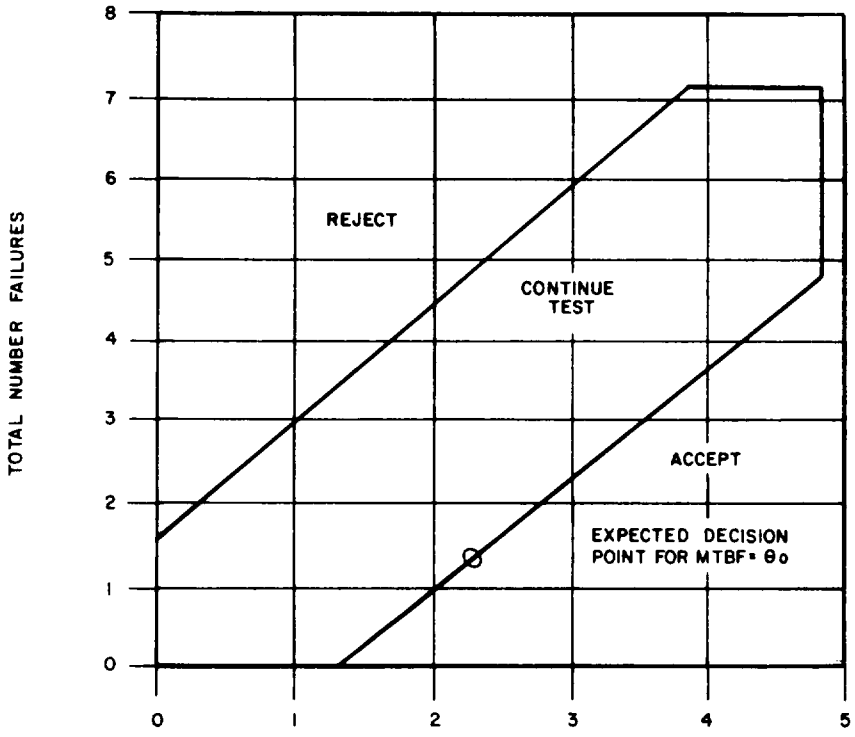
NOTE UNLESS OTHERWISE SPECIFIED ALL DIMENSIONS ARE IN INCHES

FIGURE 1. Cargo compartment profile (C-130).

X-1958

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DECISION RISKS 20%
DISCRIMINATION RATIO 2.0:1



NO. OF FAILURES	TOTAL TEST TIME * REJECT (EQUAL OR LESS)	ACCEPT EQUAL OR MORE
0	N/A	1.40
1	N/A	2.09
2	.35	2.79
3	1.04	3.48
4	1.73	4.17
5	2.43	4.87
6	3.12	4.87
7	3.81	4.87
8	4.87	N/A

*TOTAL TEST TIME IS TOTAL UNIT HOURS OF "EQUIPMENT ON" TIME AND IS EXPRESSED IN MULTIPLES OF THE SPECIFIED MTBF.

FIGURE 2. Accept-reject criteria.

X-3661A

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-D-10271H	Drilling Machine, Well, Rotary: 1,500-Foot

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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